BearingPoint. Management & Technology Consultants

Financial Services Technology Journal

The infrastructure and computing issue

Letter from the executive editor 1

This issue at a glance 4

The business of technology

Utility computing: Keys to successful implementation 6

Configuration management database: Latest developments and trends 15

Planning a pandemic response plan: Why everybody should have one 21

How it (should) all start

Data center capacity planning: An introduction and brief survey of optimization techniques 24

The new challenges of disaster recovery technology and their impact on IT infrastructure 28

Data for decision

Scenario analysis for decision making: Outsourcing versus cost takeout 34

Below the code

Managing storage capacity: Tools and techniques 39

New advances and techniques to strengthen network resiliency: A primer on new technologies 43

Summer 2007 | Volume 4 | Issue 1

Letter from the executive editor

IT infrastructure refers to "what a firm needs to make other business-specific IT resources perform reliably and efficiently." What it means today is that IT infrastructure is everywhere!

IT infrastructure is the set of technologies and services supporting robust and cost-effective access to the storage, processing and distribution of information. A few years ago, the term infrastructure was mostly used to refer to networks ("the communication infrastructure"). But, over time, organizations have come to realize that the techniques and discipline that had been used to improve the procurement, operation and life cycle management of networks could provide design and implementation patterns applicable to other cross-IT enterprise resources. Here are three examples to consider:

- **The emerging XaaS or "XoD" delivery models.**[1] In the late 1990s, "bandwidth on demand" was considered the future delivery model for communication services. The concept has been extended toward offering other types of commodity-like services across the enterprise as if they were provided by some (internal or third-party) organization functioning as a utility (including utility computing, the focus of this issue's first article). This trend also explains the simultaneous emergence of metered services and the definition of quality of services targets that are often derived from the telecommunications world.[2]

- **Catalog-based service definition.** In terms of IT offerings, communication solutions were the first that could be defined using a limited number of distinct characteristics—bandwidth, bit error rate (BER) and mean time to repair (MTTR)—that could be selected from a service catalog. The rapid provisioning of predefined server capacity with well-defined configurations selected from a service catalog is what the utility computing model promises to deliver.

- **Service management process and framework.** The IT Infrastructure Library (ITIL®)[3] is the most widely accepted approach to IT service management. ITIL applies to the entire technology stack, but it was originally developed under the auspices of the U.K. government's Central Computer and Telecommunications Agency.

[1] "X as a Service" or "X on Demand" with X ∈ {"Application", "Storage", "Computing"}.

[2] On how disaster recovery concepts such as recovery time objective (RTO) and recovery point objective (RPO) are modern cousins of the mean time to repair concept found in the telecommunications industry, see "The new challenges of disaster recovery technology and their impacts on computing" by Daryouche Behboudi in this issue of the BearingPoint *Financial Services Technology Journal*.

[3] The IT Infrastructure Library (ITIL®) is a widely accepted approach to IT service management; the ITIL has just recently been revised as ITIL Release 3.0, see http://www.itil.co.uk.

in•fra•struc•ture (in′frə struk′chər) *n.* 1. the basic, underlying framework or features of a system or organization

From its humble beginning and association with data and voice communication services, the enterprise infrastructure concept now encompasses computing and storage infrastructure (i.e., the entire data center realm). And—if we extend the original definition of infrastructure to encompass all IT resources that are used pretty much "as is" throughout the enterprise—then it becomes clear that an even greater portion of the IT stack should be considered (and managed) as an enterprise infrastructure. For example, candidate IT infrastructure also should include:

- Corporate applications such as e-mail (and possibly some enterprise resource planning systems)
- IT services (created using a service-oriented architecture and made available by several applications)
- IT security, compliance and risk management tools
- Application messaging (enterprise application integration) solutions

Last but not least, it has become apparent that a large portion of data in the enterprise should also be treated as infrastructure: the controlled but pervasive sharing of information that creates a "single version of the truth" is the condition sine qua non to produce a single view of the customer or consistent reporting of financial information.

Making a distinction between business unit-specific services and resources (data, application, computing and communication) on the one hand and those that can be shared as a common IT infrastructure on the other hand is more than an interesting exercise in taxonomy. This distinction helps leverage the lessons we learned over time in the introduction, management and evolution of infrastructure-type activities. Following are some of them, with a focus on the challenges inherently associated with the management and promotion of the IT infrastructure concepts:

- IT infrastructure should be managed at the enterprise level with a focus on supporting scalability and reduction of unit cost through economy of scale and scope.
- IT infrastructure projects are notoriously difficult to justify through a regular return on investment/business case model. Any technologist who has ever tried to convince his management to implement a service management framework or promoted a single sign-on project has endured this kind of challenge: how do you really quantify savings? The answer in many cases is: not easily.[4]
- Many stakeholders will point to the (negative) impacts on their business or area of influence when implementing a cross-enterprise infrastructure. The remedy exists: it is to make sure that these projects are sponsored at the right level to make the case based on the competitive or strategic advantage delivered at the enterprise level so that "the first business unit that needs to cross the rivers does not have to pay for the whole bridge."
- IT infrastructure technologies have a distinct life cycle: early adopters usually introduce the supporting technology outside of the usual scope of influence of the IT organization. But successful rationalization of the infrastructure is a protracted process that can only be performed if sponsored at the highest level of the organization (see the previous point).

[4]For a discussion of the relevance of business case applied to infrastructure-related projects and a discussion of alternative approaches, see "Using Real Options to Mitigate Risks in Large IT Projects" by Pierre Pureur in the May 2005 issue of the BearingPoint *Financial Services Technology Journal*.

- The pricing of IT infrastructure-based services often leads to interminable debate as to what is best for an organization. There are, however, predictable evolution patterns:
  - At first, users accepted a one-of-a-kind pricing approach to the delivery of IT infrastructure solutions. (Until the mid-1980s, point-to-point digital circuits were priced based on geographical position of the end points.)
  - They then demanded metered services. (Bandwidth on demand was all the rage in the 1990s.)
  - Eventually, unit cost became so low (thanks to growth of usage) that they demanded flat "all-you-can-eat" pricing structures. (I am sending this paper from a DSL line in Beijing, and I plan to call the United States later today using a Voice over Internet Protocol, or VoIP, service that costs a flat $25 fee per year.)

Yes, as a famous French philosopher did not say, today's discussion about chargeback for shared IT infrastructure services is just "déjà-vu all over again".

• • • • •

As we noted in the editorial section of the previous issue of this journal, IT infrastructure is the place to start to define enterprisewide architectures as a prerequisite not only to reduce costs but also to establish the foundation for harmonizing how the organization uses technology to realize the synergies across business units.

This issue of the BearingPoint *Financial Services Technical Journal* is devoted entirely to the discussion of infrastructure-related topics. One of the most innovative subjects we focus on is utility computing, an architecture style that aims at making good on the claim that computing resources can be delivered as a true commodity.

We also look at several aspects of planning, managing and operating IT infrastructure components, including network (maintaining route diversity in a communication network), data centers (analyzing cost and planning for capacity requirements), and storage infrastructure (planning for capacity). We also discuss the various financial models applicable to the management of IT infrastructure in terms of financial planning and reporting. A critical paper discusses how to build a configuration management database (CMDB) and a service catalog and how these activities are critically related.

Finally, two papers address risk management as applied to IT infrastructure, including how to develop a pandemic response plan and how to select the right technologies and metrics for a disaster recovery plan.

With a total of eight articles, all centered on IT infrastructure, this is (once again) the largest issue we have published on a single subject. Perhaps I should have disclosed early on that the executive editor started his career as a network engineer, which may have influenced the level of attention that we, at BearingPoint, bring to the overall IT infrastructure realm.

Good reading!

Manuel Barbero

Executive Editor, BearingPoint *Financial Services Technology Journal*

A personnel and personal note: This is the first issue with Laurence Thiery as our formal assistant editor. Laurence has been contributing enormously and in a number of ways to the content, development and the editing of the Journal and her new title is merely a belated but much deserved recognition of her past and continuous role.

This issue at a glance

By Laurence Thiery, assistant editor, BearingPoint's *Financial Services Technology Journal*

In past *Financial Services Technology Journal* issues we have covered infrastructure topics such as IP telephony, data center rationalization, security, email archival, call center virtualization and network architecture. However, we have never dedicated a full issue to infrastructure as a whole. That is what this issue is all about, with a focus on utility computing, not so much because utility computing is a new concept, but because the technologies that make it viable are finally maturing and financial institutions are embracing them.

IT infrastructure is such a vast domain that we will probably need another issue on such topics as the renaissance of the data center, computing and hosting patterns for computing services, a follow-on article on messaging compliance, virtualization, chargeback, new facility challenges (power and cooling solutions of today and tomorrow), and others.

For this issue, we have organized the articles within the framework used in past journals:

- **The business of technology** section explores how technology creates and delivers business value. In this section "Utility computing: Keys to successful implementation" features how financial companies' computing infrastructure can be delivered just like a commodity to address current inefficiencies and achieve economy of scale and scope.

  Many promises of using IT infrastructure optimally are not possible without a configuration management database (CMDB), and in our second article we cover the latest trends in this domain. We discuss how it is critically related to the definition of a service catalog, one of the core components of an efficient IT organization. Finally, our third article in this section is "Planning a pandemic response plan: Why everybody should have one" covers another strategic planning topic that relates directly to IT infrastructure.

- **How it (should) all start** examines the new approaches to strategy and architecture that are shaping the information technology world. This section explores how to define a disaster recovery strategy and its impact on the infrastructure. We then look at optimization techniques for data center capacity, a well-defined set of metrics and a capacity tool.

- **Data for decision** delves into the data and models needed for informed decision making. This section features an article on the various financial models applicable to the management of IT infrastructure in terms of financial planning and reporting. It also examines how to compare cost reduction initiatives and outsourcing scenarios.

- **Below the code** brings you best practices and observations on networking, computing infrastructure and other technology realms. Articles in this section help you navigate between news releases hype and the "tech-talk" that only hardcore techies can love. As a follow-on to the article "Data center capacity planning: An introduction and brief survey of optimization techniques," we explore tools and techniques to manage storage capacity. We could not resist concluding with an article on network engineering and the new techniques to strengthen network resiliency. Buckle up as we look at the three bottom layers of the OSI reference model and various technologies and advances that help enhance network resiliency.

Utility computing: Keys to successful implementation

Gabriel Leung, Pinaki Roy, Frederic Veron, Manuel Barbero

Enterprises are investing in server technology at accelerated rates. The surging demand from business users for data center capacity is adding thousands of machines annually to corporate IT environments. This demand is accompanied by related outlays for personnel, power, software licensing and space in expensive facilities.

Amidst this growth, conventional strategies for increasing data center capacity are proving appallingly inefficient. Industry server utilization rates hover in the 8 to 10 percent range (with some levels better at peak utilization time obviously, but still, overall a very inefficient use of assets) and storage utilization seldom reaches the 40 percent mark, and all this at a time when server deployment con-tinues to grow at 10 to 20 percent a year.

Another challenge is that financial service organizations often operate several "blended environments" created by mergers and consolidations. These M&A activities have created fragmented infrastructure and operations from a service and business management perspective.

The promise of utility computing is to make good on the claim (often offered, as of yet rarely delivered) that computing infrastructure could be delivered just like a commodity.

The utility computing vision is that, like providers of electricity, gas, water and other utilities, the IT organization or third-party suppliers should be able to consolidate capacity and automatically allocate resources based upon the real-time requirements of users (including application development groups, lines of business and other data center customers). This consolidation and allocation of resources is performed without calculating "peak" capacity needs and asking the IT group to build excess capacity for each application, which remains underutilized most of the time.

Configuration management database: Latest developments and trends

Zahra Rahemtulla

Managing risk, reducing downtime and the increasing demand to optimize IT operations are driving organizations to reconsider the necessity and importance of a central, accurate repository where infrastructure components required to support and provide IT services are logged, accounted for and accessible. It is no longer about having one master physical repository but having one master system of record which leverages multiple, already developed databases.

As a system of record, a configuration management database (CMDB) is critical in providing agile, efficient, measurable and cost-effective

IT services. In congruence with this, it is critical to define, document and provide an accessible list of internal IT Services to business consumers to enable internal consumers to select the services that are critical to them and to provide IT Operations with the ability to charge for services consumed.

The question often asked is "where do I start?" This article discusses the latest developments and trends to consider when selecting such a repository as well as implementation considerations associated with the design and building of an IT service catalog with an overall objective to effectively use a configuration management database to inform and support management decisions and provide service assurance.

Planning a pandemic response plan: Why everybody should have one

Gabriel Leung

A pandemic is a health emergency that has regional, national, and extreme case, global impacts. A pandemic typically impacts personnel first—people are sick or cannot report to work because areas are quarantined. Systems will be unattended if the affected data centers are inside the quarantine area. Services cannot be restored when these systems fail. A key characteristic that sets a pandemic situation apart from other disasters is that pandemics last typically more than a few months, this means that IT systems and services, particularly IT Infrastructure services are likely not to be restored for quite some time.

For some industries such as insurance, banking, and other financial services it is essential to have careful and detailed considerations of the necessary response and recovery of their operations given a pandemic event. This article provides insight into some of these considerations and overall pandemic response issues.

Data center capacity planning: An introduction and brief survey of optimization techniques

Arnaud Merlet and Takashi Yoshizaki

The activities of the financial services industry have led to rapidly increasing demands for more computing and storage resources at data centers.

This article discuss how the industry is re-engineering data centers to support this trend and how new business models are developing to measure the activity at data centers. We introduce the concept of data center capacity metrics (DCCM) to propose how companies should assess management of resources and how they should be adapted to support new demand. DCCM have to cover much more than rack density and power per-square footage. Other DCCM are provided along several different dimensions to provide practical guidelines to build efficient data centers. This article also describes other key factors in reducing unused infrastructure and overall data center operating costs. Finally, we discuss a capacity planning tool that was developed to model DCCM for planning execution purposes.

The new challenges of disaster recovery technology and their impacts on IT infrastructure

Daryouche Behboudi

In today's financial services environment, the heightened reputation and operational risk of not having a comprehensive and tested disaster recovery (DR) plan is simply too great to ignore. A dated model relying on third-party services is no longer viable when business requires recovery times of less than a day.

This article discusses various technologies that help the IT department meet the new, more aggressive business requirements.The key to an effective DR strategy is to incorporate it as a component of everything a firm does, not just as an afterthought.

Scenario analysis for decision making: Outsourcing and cost takeout

Julien Furioli and Anne Mondesert

Cost reduction efforts are usually set in a tactical and often annual cycle, whereas outsourcing initiatives take place over a longer-term, multiyear time frame. In this article, we use the real-life example of a leading financial services firm to illustrate how conducting a cost-savings investigation does not preclude assessing outsourcing options and taking advantage of the momentum.

We examine how large IT organizations in the financial services industry compare cost reduction initiatives and outsourcing scenarios and set up financial models and develop specific cost simulations. We also examine some of the benefits and limitations of available financial models.

Managing storage capacity: Tools and techniques

Arnaud Merlet

Risk mitigation, business continuity requirements and digitization, as well as record-keeping requirements, are driving the need for extensive storage space. In addition, much, if not all, business-critical and regulated data needs to be mirrored, replicated, backed up, archived and retained. These new requirements call for even more disks and tapes.

Nevertheless, as a result of poor capacity management processes, improper tier allocation, confusion over storage metrics, uninformed purchase decisions and outdated backup policies, a lot of storage space is wasted. This article offers seven techniques to assist in dealing with the exponential growth of raw data that most organizations are facing today, as well as an overview of a storage capacity planning tool developed by BearingPoint.

New advances and techniques to strengthen network resiliency: A primer on new technologies

Rajiv Jain

Unpredictable issues in the global telecommunications grid can have a serious impact on business operations. Yet users take for granted that service will be always available.

Although the quality of services now available from telecommunications carriers is radically better than it was a mere one or two decades ago, enterprises are now completely dependent on their communication networks to function to a degree that was unimaginable in years past.

This article presents recent advances and existing technologies that have emerged as the essential building blocks for implementing added network resiliency. We also consider the budgetary considerations associated with the adoption of these techniques. ■

The business of technology

Utility computing: Keys to successful implementation

Abstract

Abstracting the software and operating system (OS) layers in the computing infrastructure stack through virtualization techniques is not a new concept. But beyond (and building on) virtualization, Utility computing is an architecture style that aims at making good on the claim that computing resources can be delivered as a true commodity.

The technologies and frameworks that make utility computing viable are finally maturing, and this paper provides an introduction to the concepts and issues that need to be considered in evaluating the UC paradigm and emerging point solutions and frameworks. Following a brief introduction to the fundamental concepts and a maturity model attached to utility computing, we propose insights on how to develop a high-level functional utility computing architecture. We also cover some guidelines on how to develop a utility computing business case and offer a description of the key success factors to consider and the challenges to address by organizations hoping to capture the benefits of utility computing. Among those are the ability to increase server utilization rates, reduce the requirement to build capacity, and lower operating costs in data centers and across the enterprise.

Introduction: The promise of utility computing

Enterprises are investing in server technology at accelerated rates. The surging demand from business users for data center capacity is adding thousands of machines annually to corporate IT environments. This demand is accompanied by related outlays for personnel, power, software licensing and space in expensive facilities.

Amidst this growth, conventional strategies for increasing data center capacity are proving appallingly inefficient. Industry server utilization rates hover in the 8 to 10 percent range (with some peak time, obviously, but still overall a very inefficient use of assets), and storage utilization seldom reaches the 40 percent mark—all this at a time when server deployment continues to grow at 10 to 20 percent a year.[1]

Another challenge is that financial service organizations often operate several "blended environments" created by mergers and consolidations. These have created fragmented infrastructure and operations from a service and business management perspective.

The promise of utility computing is to make good on the claim (often offered, as of yet rarely delivered) that computing infrastructure could be delivered just like a commodity.

The utility computing vision is that, like providers of electricity, gas, water and other utilities, the IT organization or third-party suppliers should be able to consolidate capacity and automatically allocate resources based on the real-time requirements of users (including application development groups, lines of business and other data center customers). This consolidation and allocation of resources is performed without calculating "peak" capacity needs and asking the IT group to build excess capacity for each application, which remains underutilized most of the time.

Utility computing paradigm

Implementing the utility computing promise is about implementing a computing environment that provides on-demand computing as a general-purpose infrastructure delivered over the network to all applications and users on a subscription-fee/service level basis.

"On-demand computing" means that resources are mostly, if not completely, automatically added and removed from the environment based on workload and performance characteristics. In the mainframe and to some extent the midrange space, a number of financial engineering models have been implemented to approximate this notion. But, at least in the midrange environment, they still require significant manual intervention to deliver on-demand capacity to

The promise of utility computing is to make good on the claim (often offered, as of yet rarely delivered) that computing infrastructure could be delivered just like a commodity.

[1]Source: BearingPoint research and analysis of existing clients.

Implementing the utility computing promise is about implementing a computing environment that provides on-demand computing as a general-purpose infrastructure delivered over the network to all applications and users on a subscription-fee/service level basis.

any given application. With the advent of better workload management and provisioning technologies, this now can be done in quasi-real time across a number of different platforms.

"General-purpose infrastructure" denotes the desire to separate the software layers from the hardware layers so that increased flexibility, and therefore higher efficiency, can be reached. Platform or stack standardization becomes a recognized benefit of, but is not a prerequisite for, the enterprise and helps converge today's separate environments, regardless of their utility computing maturity. A well-architected and instrumented utility can accelerate that convergence.

"Delivered over the network," though commonplace nowadays in corporate environments, stands for the ability to provision resources (e.g., CPU, storage, etc.), regardless of their location, for the user community at the lowest cost, while still complying with minimum performance requirements. Other aspects of leveraging this architecture for utility computing bring additional benefits, including but not limited to a self-healing environment and inherent support of disaster recovery and business continuity requirements.

"Subscription-fee/service level basis" represents how utility computing allows business users to subscribe to the service they need, and only the service they need, in accordance with its level of affordability. This gives them the ability to make informed decisions based on their business requirements, rather than attempting to specify the hardware solutions that should not matter to them as long as service levels are adequate and costs are acceptable. Without having to know how this utility computing is configured, users can subscribe to different levels of service depending on their business needs (volume, performance, growth, etc.) and how much they can or want to invest. And they will pay only for that to which they subscribe.

From the user's standpoint, the benefits of utility computing include:

- **Simplifying the environment.** The computing environment can offer a standard interface for easy and fast access to the available capacity of the utility's resources.
- **Transparent measurement and billing of usage.** The utility computing model makes it possible to measure and report on usage and to bill based on that usage.
- **Well-defined services with options.** Utility computing gives access to well-defined services with options for service levels and other characteristics. A complete set of discrete computing services with different characteristics and price points can be offered.

From the IT organization's perspective, this means:

- **Implementing a new operating model.** Principally impacted are the capacity, configuration, change and service level management processes.
- **Developing a new financial model.** Resource usage accounting is done on a per-service basis, not just on a per-box basis. This approach facilitates detailed chargeback and billing for more accurate accounting and management.
- **Leveraging the new technology paradigm.** This includes:
  - Dynamic and automatic resource provisioning, unifying all computing and storage resources in a single framework.
  - Services provisioning management, automating software and hardware installation and configuration.
  - Policy automation, creating the ability to manage the allocation of IT resources dynamically based on predefined business policies and service level objectives that are consistently maintained in dynamic business environments.

While all these notions have been at the center of the strategy and vision of most IT organizations over the past few decades (with the appearance of distributed computing), they have only been accomplished in the mainframe world. With the maturation of new technologies and process frameworks, IT organizations are now positioned to deliver on these promises.

Utility computing versus virtualization

Over time, IT practitioners and vendors have devised a number of approaches, techniques and architectures to better use computing and storage resources. Techniques have included partitioning (mainframe and midrange), clustering, virtualization and grid computing. Out of this list of techniques, virtualization is often mistaken as a utility computing solution or as an absolute prerequisite to a utility computing architecture. These misunderstandings lead IT professionals into debates that limit their ability to deliver on the promises of both paradigms.

To clarify our understanding, we propose to look at the two major differences between utility computing and virtualization. These two concepts are related because, while utility computing builds on the principles and features of virtualization, it does not require complete virtualization either:

- **Utility computing focuses on how service is delivered.** For most organizations, the real challenge resides not so much in getting servers to run several operating systems concurrently in a way that is transparent to users (virtualization technology has delivered that paradigm), but in introducing and supporting the management process that makes possible automatic configuration and service level management, as well as computing asset provisioning and resource management. Utility computing goes beyond virtualization in that it stresses the overall "service management" components of the promised IT commodity.

- **Utility computing's scope is about dynamic provisioning.** Classic virtualization techniques are very good at improving server densification. Densification is about using the same servers for multiple instances of different applications and operating systems. Application and operating system images are pre-imaged to form a fixed instance.

But, in this model, servers either are up or not (under load or idle). Therefore, it is not possible to add servers dynamically. Moreover, servers (and storage) capacity cannot be repurposed for other uses while idle. Application and operating system combinations are essentially fixed and cannot be easily changed.

Utility computing supports dynamic provisioning, the automated and fast delivery of new resources configured to users' specifications. Application and operating system images are maintained in a data store, and the images can be dynamically loaded or unloaded based on predefined policies. The same server can be used for other purposes while it is not needed (e.g., the servers can be repurposed for development or testing).

Another way of looking at the difference between the two approaches is to note that virtualization is mostly about optimizing the configuration of a physical device to better share it across the workload and keeping the different workloads protected from one another. Utility computing is about how to better manage the overall environment, making sure that all resources are configured the way they should be, deliver the appropriate level of resources (virtual or not) to the applications, and report on how the overall asset base is used. This requires more than the classic "hypervisor" tool that virtualization solutions offer.

Figure 1 represents the technology view of our utility computing framework in which virtualization is often part of the solution. Utility computing incorporates additional notions and technologies to form a whole aimed at improving overall computing efficiency and manageability.

A proposed maturity model

As with many IT paradigms (e.g., object-oriented programming and service-oriented architecture), utility computing is an architecture style (Figure 1). Its degree of adoption by an organization should be planned and discussed using a maturity model similar in spirit to those used in other domains. (The five levels of the capability maturity model come to mind.)

Figure 2 identifies such a model applied to utility computing (and related governance and business process enablers). The model defines, for example, level 3 as the "virtualized" stage, at which software and hardware platforms have been through further optimization, but this is by no means a necessary step for all machines in the utility.

We have found the maturity model useful in discussing the utility computing concept when it has been mapped back to business value. The higher the maturity level, the more the utility computing is aligned with the business, hence its value. For example:

- **Level 1 (cognizant)** is the level at which an organization has inventoried its computing assets. This alone should help create value by identifying depreciated requirements and redundant systems or licenses.

- **Level 2 (instrumented)** describes the stage at which an organization has introduced tools and processes to manage its IT computing infrastructure proactively, thus lowering the cost of operation and/or increasing service quality or security monitoring.

Utility maturity levels can be also used to describe how key aspects (views[2]) of the IT environment will evolve over time (Figure 3), including:

- How technology will be used.

- How processes are redesigned.

- What service will be delivered.

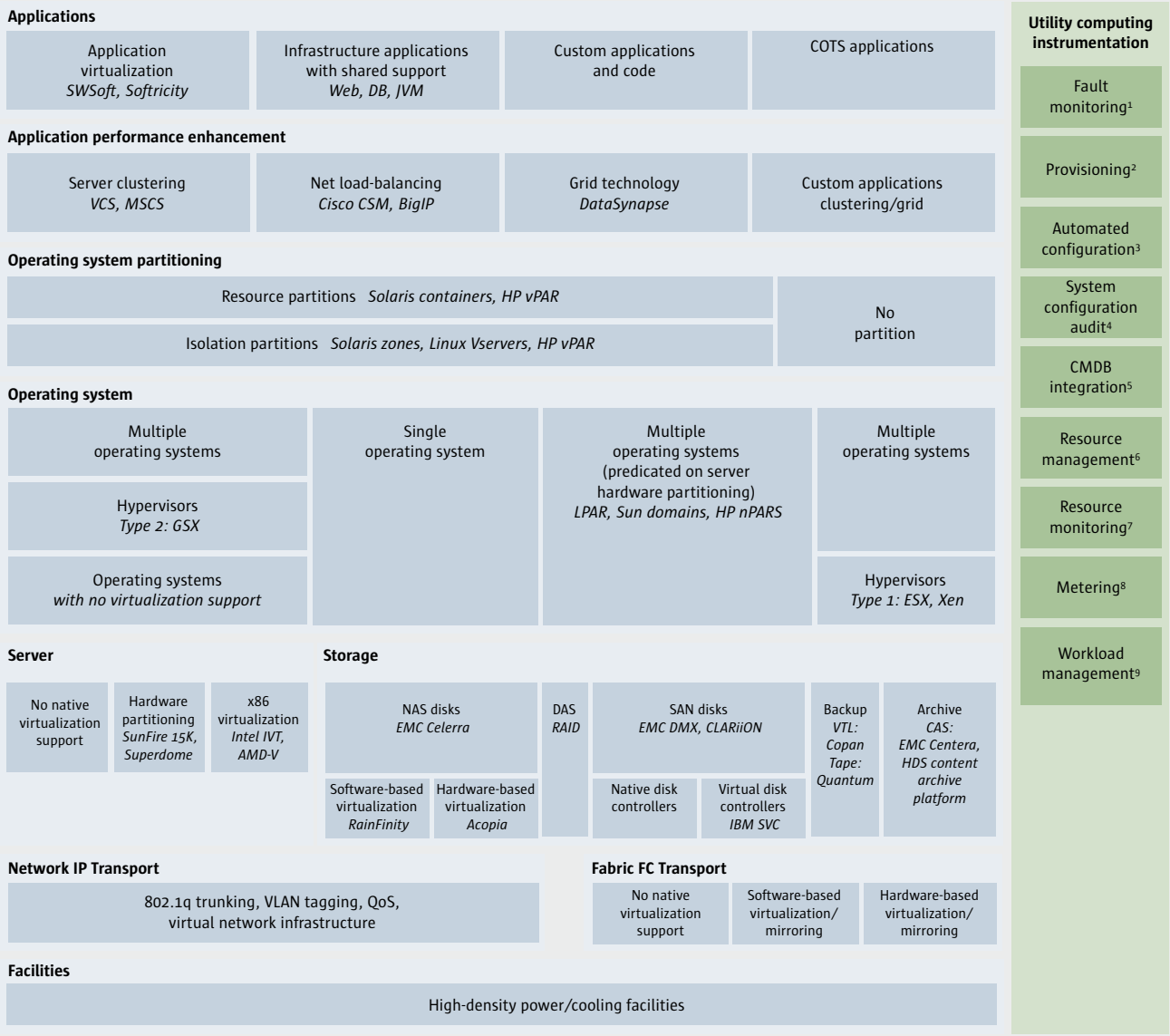
- What skills people will acquire.

- How the governance model will develop.

For example, an organization targeting level 4 will have to make sure that, from the technology standpoint, its environment is fairly standardized (e.g., only a limited number of operating systems and versions are supported) and that, from the people standpoint, the processes are in place to repeat earlier successes with similar services with similar service levels (i.e., repeatable, automated 3).

[2]"View" and other key architectural concepts are used throughout this paper. For a formal definition and nomenclature for these terms, see "Architecting the IT organization on |arch-itecture tools and techniques and their application to (IT) organizational design" by M. Barbero in this issue of the BearingPoint *Financial Services Technology Journal*.

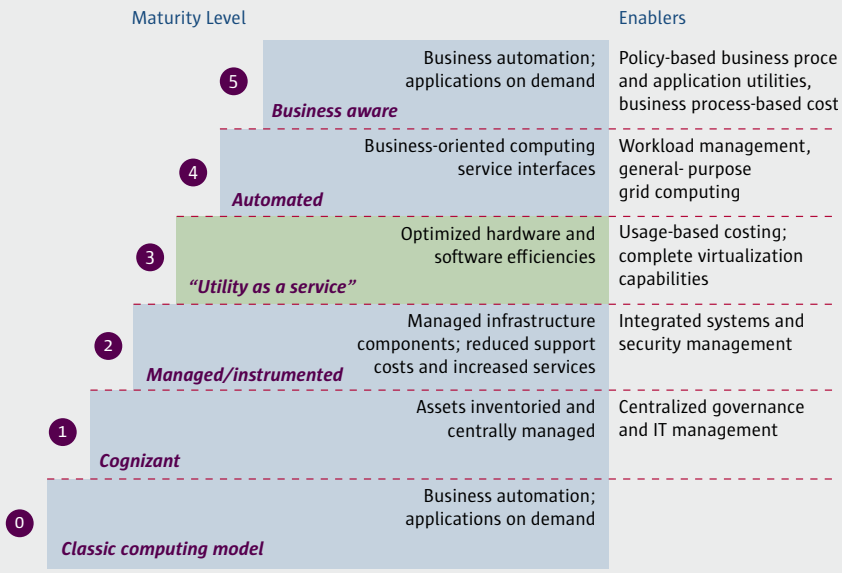
Figure 1. Technology view of the utility computing framework



Notes:
[1]Monitoring of virtual instances.
[2]Includes bare-metal, operating system upgrade, patch, virtual machine, application.
[3]Alteration of content/configuration files/registry entries across multiple systems simultaneously (NOT automated remediation).
[4]Greater system policy adherence required to ensure "free pool" of resources is accurate.
[5]Enhanced configuration records, especially VMs.
[6]Management of resources (e.g., CPU, memory, network I/O, disk I/O).
[7]Utilization monitoring.
[8]Metering separated as may be needed based on number of VMs initially rather than usage.
[9]Dynamic addition of resources to applications as needed through automated provisioning (horizontal scaling)/automated resource management (vertical scaling)/automated application migration.

Virtualization is mostly about optimizing the configuration of a physical device to better share it across the workload and keeping the different workloads protected from one another. Utility computing is about how to better manage the overall environment, making sure that all resources are configured the way they should be, deliver the appropriate level of resources (virtual or not) to the applications, and report on how the overall asset base is used.

Figure 2. A utility computing maturity model



This maturity model assumes that progress toward achieving Maturity Level 5 comes in stages.

The model is used to determine the maturity across all components of the foundational framework.

Maturity progression is evolutionary and may not be equal for each framework component since achieving maturity at each level is used to build a foundation upon which to achieve the subsequent level.

Different weighting and maturity criteria will apply to the different components at each level (maturity will vary) This should enable focus on less mature, less structure areas. This model can also serve as a continuous improvement tool in the process of moving from as-is maturity to to-be maturity.

A functional reference architecture

The utility computing model requires the implementation of a stack of services to configure and then allocate computing and, in many cases, storage resources. These services are automated in each of the configuration and allocation domains as depicted in Figure 4.[3]

The proposed functional reference model for utility computing relies on the existence of a configuration management database (CMDB)[4] to facilitate configuration management and then support other classic services of the IT Infrastructure Library (ITIL®) model[5], including asset, incident, problem, change and release management.

Once the IT organization has acquired the ability to manage and monitor the underlying virtualized computing assets proactively, an automatic resource provisioning framework is needed. This framework isrequired to define and implement the various processes that unify computing resources into a single resource pool that can be managed to respond automatically to the changing capacity requirements of applications in real time.

Functions supported in the provisioning framework include resource discovery, application packaging (before implementation) and the other functions described in Figure 4.

At that level, another set of functions can be introduced (not depicted in Figure 4). Both allow data centers to perform "hands-off" provisioning of servers with the appropriate operating systems and applications:

- Policy automation helps enterprises dynamically manage the allocation of IT resources to applications based on predefined business policies and service level objectives.
- Service provisioning management allows the automatic installation and configuration of new hardware and software when an application needs additional capacity.

Bringing these components together allows the utility computing infrastructure to monitor the varying performance levels of an application and determine—based on a predetermined policy—when that application needs additional computing power. If it does, a request for provisioning is sent to the system, which automatically images the server, brings it up to speed and makes it available to the application.

Devising the utility computing logical and physical architecture

After years of promises—and even hype—the technologies that support utility computing are being brought to market by established vendors, as well as by smaller industry-leading firms.

For designing a solution and selecting vendors, the utility computing maturity model is useful to assess each component of the architecture in terms of supporting functionality and the delivery model along the several views of the technology stack (network, server, operating system, partitioning and applications, where the latter can be understood as how application performance can be improved).

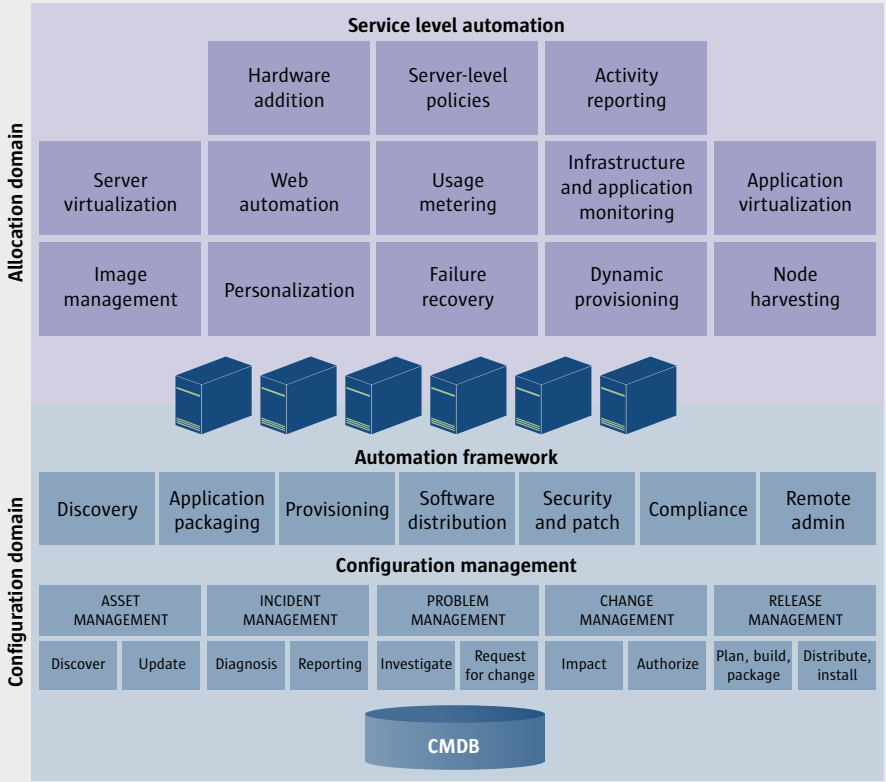
The utility computing maturity model helps ascertain how the point solution and a framework provided by vendors are capable of bringing the utility computing environment to a given maturity level. For example (Figure 5), to achieve a utility computing maturity level 4 (automated), the majority of hardware used in the data centers needs to support virtualization natively.

[3]Note that this brief introduction to a proposed functional architecture is deliberately not exhaustive since it would be fine-tuned based on the stated goals of the utility computing program within a given enterprise.
[4]See the article, "Configuration management database: Latest development and trends," in this issue of the BearingPoint *Financial Services Technology Journal*.
[5]The IT Infrastructure Library (ITIL®) is a widely accepted approach to IT service management; see http://www.itil.co.uk.

Figure 3. Defining the level of penetration across key architectural views with the utility computing maturity model

Utility maturity levels	Technology view	Process view	Service view	People view	Governance
HIGH 5 Business aware — Advanced optimization and environment agility	Value through business demand/capacity management Standardized technology environment	Process management integrated with key business functions, aligned to standard stacks and automated service provisioning	Integrated and automated with business demand that is managed by a service	Organizational model optimized and aligned with key functional areas; actionable enterprisewide process knowledge	Continuous monitoring through specific KPIs Continual feedback and feed forward controls
4 Automated — Optimization execution	Service-driven technology provisioning	Systemic, defined automated processes (capacity, problem, incident, configuration)	Discipline in place to repeat earlier successes on similar services with similar service levels; automated service provisioning	Discipline in place to repeat earlier successes on similar services with similar service levels (repeatable, automated enablement)	Predictable and quantifiable service level measures
3 Utility as a service — Implementation of shared services concepts and initiate alignment with user group requirements	Reactive technology demand/capacity management	Service-driven utility-based provisioning	Services aligned with enterprise chargeback models; service catalog in place	Services aligned with enterprise chargeback models	Integration of metrics into service life cycle management
2 Managed — Implementation of foundational elements	Little or no technology integration and/or standardization (heterogeneous environment)	Reactive process management with discrete functional domains	Standard services are defined	Standard services centrally managed (manual process)	Metrics defined but not centrally recognized, enforced or integrated
1 Cognizant — Acceptance and call for action	Lack of virtualization	Reliance on manual or labor-intensive process management activities	Little or no enterprisewide awareness	Standard services are defined	Lack of defined process/organization/technology/service metrics
LOW 0 Classic — Heterogeneous, non-optimized computing environment		Disparate processes		Little or no enterprisewide awareness	No framework for orderly improvement

Figure 4. Proposed functional architecture for utility computing



The proposed functional reference model for utility computing relies on the existence of a configuration management database (CMDB) to facilitate configuration management and then support other classic services of the IT Infrastructure Library (ITIL®) model, including asset, incident, problem, change and release management.

The proposed utility computing maturity model can be combined with the technology view of the utility computing framework to identify how a given product can support each function described in the reference architecture model.

In our experience, the current state of the art still requires the integration of various point solutions since no single integrated solution can currently support the whole set of utility computing functions across all levels of the maturity model.

Utility computing benefits: Building the business case

Developing a credible business case for any infrastructure building project is always a delicate endeavor[6] that often requires some creativity. It is relatively easy to quantify implementation costs while identifying actual cost savings that are not merely "cost avoidance."[7]

That said, the situation for utility computing is relatively easier; since the utility computing system is automated, it doesn't require human intervention. Automating the provisioning, configuration and management of the computer processing allocation process facilitates a significant reduction in the number of personnel hours required to run the data center.

Figure 5. Identifying the characteristics of the technology stack using the utility computing maturity model

Utility maturity levels (HIGH → LOW)	Network	Server	Operating system	Partitioning	Application performance enhancement
5 Business aware — Advanced optimization and environment agility	LAN QoS allowing different quality of service levels for each IP address	Only hardware that supports virtualization natively (i.e., hardware partitioning or x86 acceleration in data centers)	Preferred virtualization methods have been chosen and only ones allowed in data center Beyond virtualization, application sharing of an OS is normal	Complete namespace and resource isolation exists in all utility environments Allocation of new resources as required on an as-needed basis	Use of high-availability technologies as standard rollout Applications designed to scale horizontally not vertically
4 Automated — Optimization execution	Use of technologies to allow IP addresses to failover between subnets seamlessly	Majority of hardware in data centers supports virtualization natively	Some non-preferred variants of virtualization Preferred form of virtualization chosen and used in new implementations	Resource partitioning of CPU, memory, network I/O, disk I/O used on all new servers in data center	Grid products used consistently for application scalability
3 Utility as a service — Implementation of shared services concepts and initiate alignment with user group requirements	VLAN trunking directly to server to allow multiple VLANs per server	All new hardware purchased supports virtualization natively	Most forms of virtualization available allowing greater utilization of hardware and application flexibility Ideas around preferred methods going forward	Some implementations of resource partitioning Namespace separation always implemented	Native application clustering technologies used (e.g., Oracle® RAC)
2 Managed — Implementation of foundational elements	NIC-teaming for true redundancy and increased throughput	Some platforms with x86 virtualization support	A number of virtualization technologies exist No cohesive virtualization plan, just capability	Some namespace separation through various technologies	Server clustering technologies used consistently to provide HA
1 Cognizant — Acceptance and call for action	Multiple NICs per server allowing failover and multiple paths for redundancy	Some platforms with support for hardware partitioning	Some virtualization as an emerging technology	Namespace separation only through use of a virtual OS	HA provided consistently for Web layer through network load-balancers
0 Classic — Heterogeneous, non-optimized computing environment	Traditional network connectivity with one LAN connection per server	Traditional server hardware with no attention paid to native virtualization support	Traditional model of 1 OS per physical server No virtualization exists	Traditional model with little application sharing Where sharing does exist, no namespace/resource isolation	Occasional use of HA technologies as required on a per-application basis No real application scalability

[6]See a discussion of the relevance of business case applied to an infrastructure-related project and a discussion of alternative approaches in "Using real options to mitigate risks in large IT projects" by Pierre Pureur in the May 2005 issue of the BearingPoint *Financial Services Technology Journal*.

[7]Any technologist who has ever tried to convince his management to implement a service management framework or some sort of single sign-on project has endured this kind of challenge: how do you really quantify saving? The answer in many cases is: not easily.

Thus, it frees up data center resources to be either reduced or redirected to work on other higher-priority tasks. Additionally, the processing resources that are "on call" don't cost anything (or very little and definitely less than the under-utilized resources in today's raised floors) until they are turned on. In addition, tangible cost savings can be derived through:.

- **Reduced number of physical servers required.** Getting more out of fewer assets dramatically reduces corporate investment in data center hardware.
- **Decreased software license fees.** Lowering hardware requirements significantly reduces software licensing needs.
- **Decreased facility space requirements.** Fewer physical machines require less floor space, making facilities available for other IT functions or for corporate use.
- **Reduced power consumption.** Fewer servers mean less energy required for running and cooling hardware. Nowadays, power utilities provide organizations adopting these types of solutions with additional discounts on power costs.

Although one of the main goals of implementing utility computing is cost reduction, organizations can achieve other benefits that should be factored in the business case, such as the returns a business can derive from moving to this next-generation computing environment. These are difficult to quantify, but are critical nonetheless, and include:

- **Facilitated disaster recovery.** Transforming the data recovery environment from a physical to a virtual one can reduce or eliminate the need for physical servers waiting offline in case of emergency. Instead, if a disaster occurs, sufficient capacity in the virtual "pool" of process-ing capacity allows organizations to replace inoperable resources quickly with viable ones and get users online in a more timely manner.
- **Faster time to market for new IT applications and services.** During application development, data center users benefit from their ability to "play" with computing resources when deciding which features and technologies to implement. Additionally, if the performance of online applications decreases due to a surge in usage, automatic provisioning can add resources to address the increased demand in real time. Utility computing facilitates both of these capabilities. It also supports faster time to market for internal IT projects and safeguards those applications strategic to the organization, customers and partners against performance degradation.
- **Greater risk mitigation.** Well-defined policies manage the use of a "self-healing" architecture and a less complex standardized computing stack.
- **Improved compliance.** The configuration management domain allows for quick identification and resolution of configuration drifts and outdated versions and patches with well-defined control points that are highly integrated with automated operations, allowing for greater flexibility to monitor process execution.

Some IT organizations are projecting utilization rates as high as 80 percent and a 45 percent cost reduction through the implementation of the latest utility computing models and solutions.

Against these cost savings and other benefits, one needs to consider the hidden costs associated with the implementation of utility com- puting beyond the cost of the new technology components. In particular, these estimates need to include the cost of addressing five categories of challenges in implementing utility computing, which include:

- **People.** As with any major organizational transformation, various stakeholders will resist change. Users may be suspicious that their costs will increase or their access to processing capability will diminish. Data center workers may worry about being displaced. Additionally, the shift to utility computing will require development of training and related documentation for data center employees and users alike. Mitigating these risks requires commitment to allocating sufficient budget to support change management and communication efforts.

Developing a credible business case for any infrastructure building project is always a delicate endeavor that often requires some creativity. It is relatively easy to quantify implementation costs while identifying actual cost savings that are not merely "cost avoidance."

- **Financials.** Because utility computing is such a new operating model for data centers, the financial function in many organizations may not be set up to manage it effectively. The usage-based billing and chargeback processes that are critical aspects of utility computing, in particular, require significant changes to accounting procedures. Additionally, many vendors have yet to offer licensing agreements that accommodate virtualized environments, making it necessary to renegotiate and redesign existing contracts at additional cost.
- **Processes.** Data centers will need to create standardized builds for policy deployment and enforcement throughout the operations life cycle. They will also need to implement change management procedures as key processes are re-engineered.
- **Technology.** Coordinating all the different technology components and managing the many vendors involved in a utility computing initiative can be difficult. IT management must be able to apply policies consistently across all server, storage and network domains. Capacity planning and service level agreements will become more critical as server provisioning transitions from a months-long process to one that takes minutes.

- **Application remediation.** Ideally, the application portfolio should be able to run in the new environment without changes. But that position assumes that all currently supported hardware and operating system software combinations will continue to be supported in the new environment, something that might not be the case as the organization deliberately chooses to move toward a limited number of supported configurations. Moreover, merely conducting an analysis of the current application portfolio and developing a transition (or remediation) plan to migrate each application to the utility computing environment is also the source of additional costs (and risks).

Overall, there are too few full-fledged implementations of utility computing to identify a statistically accurate estimate of return on investment, but organizations currently implementing utility computing at the proof-of-concept level have developed realistic financial models that take into account the key parameters described above in order to achieve relatively long (i.e., more than 24 months) payback objectives.

Requirements for success

Organizations considering implementing utility computing must meet a number of key requirements to reach their cost targets and business goals. The success factors we have identified include:

- **Putting the right processes in place.** Since organizations are automating the processes for adding infrastructure capacity, they must make sure that those processes actually work, and work efficiently. Unlike manual capacity upgrades, data center personnel won't have the option of "tweaking" systems and handling exceptions as they arise.

- **Creating a fair, transparent and financially sustainable chargeback model.** Since utility computing allows data center customers to share computing power independently of its physical source, organizations need an accurate way to measure what users consume and charge them for it appropriately. This dramatically changes the way that the IT organi-zation recoups the cost of its infrastructure. It is therefore essential to create a charge- back model that precisely reflects utilization of data center resources by customers while covering all capital and operational costs.

- **Standardizing the "outlet" or computing stack.** Like other utilities such as gas, electric and water services, the utility computing environment needs a standard interface that allows users to access the processing resources they require quickly and easily. This means standardizing server imaging builds into a manageable, well-defined set of possible configurations. Also, a functional utility computing environment should provide users with a catalog of services offering different characteristics, options and related price points from which they can choose what they need.

- **Streamlining organizational relationships.** Similar to provisioning users with physical servers, requests for additional capacity must be fulfilled according to predefined service level agreements. However, in a utility computing environment, the IT group usually has only hours, rather than weeks or months, to accomplish this. As a result, interactions between users and the data center team must change substantially, requiring dramatic redefinition of organizational relationships.

- **Implementing a pilot that demonstrates the value of utility computing to all stakeholders.** At first, many data center customers may resist the move to utility computing. They worry that they are getting less power for their money and feel constrained by being forced to choose from a predefined menu of options. A pilot project that transitions those applications to the utility model with the highest chance of success can demonstrate savings and operational efficiencies early in the process and establish momentum for the project going forward.

Applying any one of these recommendations is, in and by itself, beneficial in any environment or in the context of any fundamental change affecting the IT group. To that extent, introducing utility computing is also an opportunity to implement leading practices in each of these areas.

Conclusion

This article has provided a concise introduction to utility computing that we hope will help decision makers and IT practitioners envision the scope, key concepts and issues that need to be considered in evaluating the utility computing paradigm and individual point solutions and frameworks within a proposed maturity model. Organizations that use Design for Six Sigma (DFSS) as an approach will recognize the elements of the define, analyze and select phases of the process.

Overall, new technological advances are making utility computing a viable proposition for enterprises. A dramatically improved way to acquire, allocate and charge for use of data center resources, utility computing can reduce and better monitor IT expenses and the velocity of their growth while offering significant improvements in operational efficiencies.

At the enterprise level, utility computing provides an effective way for data centers to scale to meet ever-changing needs for processing capacity. As organizations increasingly depend on IT resources to compete successfully, utility computing can improve their ability to respond agilely to emerging market challenges, provided that significant attention is given to the non-technical aspects of the issues at stake.■

Configuration management database: Latest developments and trends

Abstract

The configuration management database (CMDB) is a critical system of record in providing agile, efficient, measurable and cost-effective services. It also is essential to define, document and provide an IT service catalog accessible to both internal and external consumers.

The question often asked is "where do I start?" This article discusses the latest developments and trends to consider when selecting a CMDB, as well as considerations when designing and building an IT service catalog with an overall objective to effectively use the information to drive management decisions and provide service assurance.

Overview

Managing risk, reducing downtime and the increasing demand to improve IT operations are driving organizations to reconsider the necessity and importance of a central, accurate repository of key information about the environment. Such a resource is not just a physical master repository, but a master system of record that leverages multiple, established databases. It is a place where infrastructure components required to support and provide IT services are logged, accounted for and accessible.

This system of record is critical in providing agile, efficient, measurable and cost-effective IT services.

At the same time, many organizations have committed to defining, documenting and providing an accessible, internal IT service catalog to business consumers. This will help internal consumers select the services that are critical to them and provide IT operations with the ability to charge for services consumed.

The configuration management database (CMDB) concept helps create a central repository for storing basic records related to configuration items (CIs)—including servers, routers, desktops, business processes and licensing information—in an organization's production environment and define the relationship among those items.

The IT Infrastructure Library (ITIL®)[1] defines configuration management as a process consisting of the following:

- **Identification:** specifying and identifying all IT components or CIs and including them in the CMDB.
- **Control:** managing each CI, specifying who is authorized to "change" it.
- **Status:** recording of the status of all CIs in the CMDB and maintaining this information.
- **Verification:** reviewing and auditing to verify that the information contained in the CMDB is accurate.

The CMDB can be viewed as an "enabler" of the configuration management process. It is the central repository that provides IT managers the capability to access an integrated set of data about the CIs in their production environment. With it, IT managers are able to make more informed decisions on the impact and risk of any infrastructure changes to a service that the business actually consumes.

As simple as the concept may seem, however, the CMDB has remained a somewhat confusing and abstract notion for many organizations that have opted to implement the ITIL framework to help them define, deliver, manage, optimize and measure IT services.

The CMDB is not just a physical master repository, but a master system of record that leverages multiple, established databases. It is a place where infrastructure components required to support and provide IT services are logged, accounted for and accessible.

[1]The IT Infrastructure Library (ITIL®) is a widely accepted approach to IT service management; see http://www.itil.co.uk.

Before considering a CMDB deployment, IT leaders must ask the following key questions:

- Do IT operations align with business needs?
- What impacts operational effectiveness? For example:
  - Failed or unplanned changes?
  - Loss of productivity?
  - Service disruption to the business?
- What do I want to achieve from both an IT and a business perspective?
- How will this demonstrate value to my business?
- Will implementing a CMDB provide a return on investment?
- Does my organization have the skill level and resources to support and maintain a CMDB?

Some trends in CMDB solutions

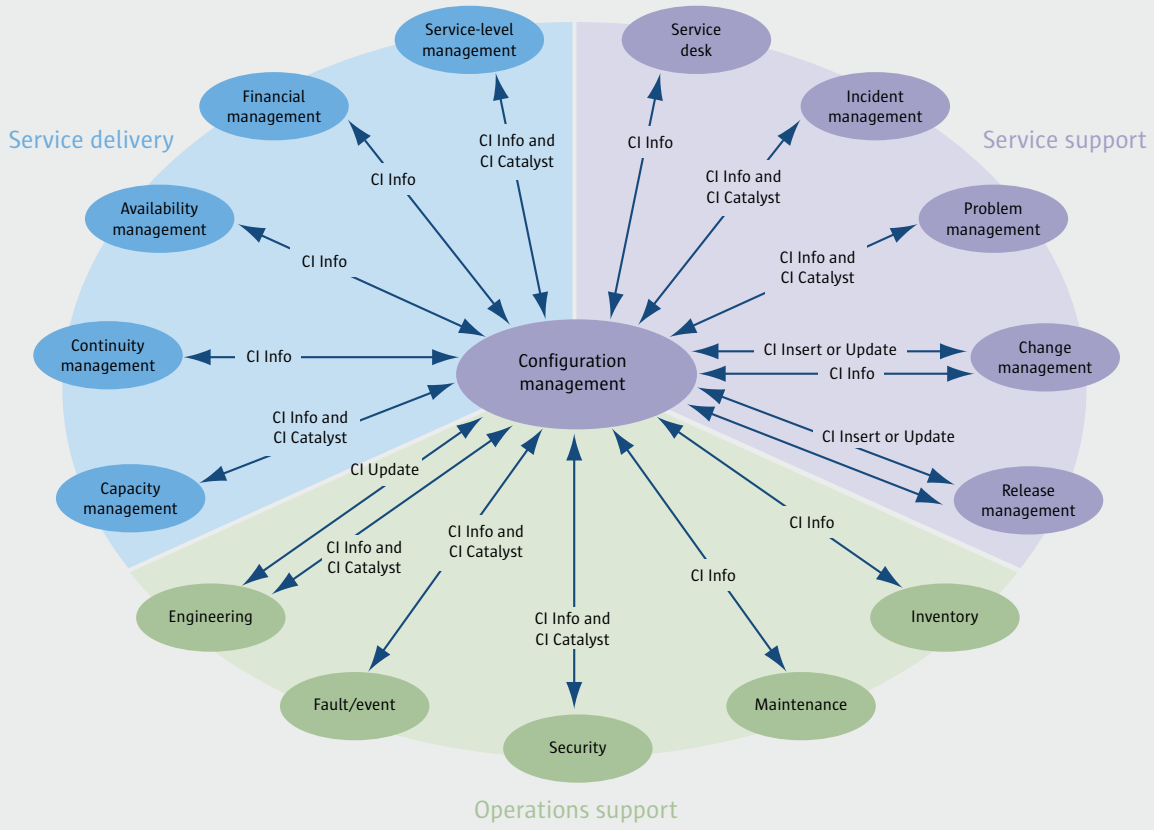
The CMDB marketplace is still considered an emerging market. CMDB vendors are positioning solutions that will allow for effective and maintainable implementations. Organizations should not forget that the CMDB is part of the ITIL-based configuration management process and that one part of the process cannot stand alone. Keys to selecting a CMDB solution are understanding and defining any process interdependencies before choosing a CMDB tool. Once these interdependencies are defined, an organization should create a realistic implementation road map and select tools based on specific goals and organization objectives.

When deciding where to begin a CMDB deployment, the following 10 steps are recommended:

- **Step 1:** Obtain resource and budget commitment. Often, this element ultimately determines the success or failure of a CMDB deployment.
- **Step 2:** Assess the current maturity of the configuration management process.
- **Step 3:** Define an end-to-end process and tool architecture with clear interrelationships that are critical to managing and sustaining a CMDB. Some key interrelated processes are:
  - Incident management
  - Problem management
  - Change management
  - Availability management
  - Service level management

  The process relationships with configuration management as the "nucleus" are depicted in Figure 1.
- **Step 4:** Determine if the business model supports a centralized or federated CMDB model (Table 1).

Figure 1. Configuration management at the center of operations



- **Step 5:** Assemble the project team and define the project. Verify that operational resources are also identified and that process ownership is clear.
- **Step 6:** Define requirements, such as:
  - Data management and reconciliation
  - Integration of information
  - Mapping of CI relationships
  - CI correlation data
  - Application dependency
  - Scalability
- **Step 7:** Finalize requirements, evaluate the vendors and select technology.
- **Step 8:** Construct and deploy the CMDB.
- **Step 9:** Communicate ongoing value and drive continuous improvement through ongoing maturity assessments, compliance audits and governance structure.
- **Step 10:** Periodically revise the architecture, process and business value delivered by the CMDB.

Table 1. Centralized versus federated CMDB

Centralized CMDB	Federated CMDB
• Can be considered to exist as one physical database. • Usually not recommended for the large, complex IT environments of global financial services organizations due to the complexity of creating and maintaining one large database with multiple sources of data that exist in a large organization.	• *Federation* refers to a central repository holding CIs and their relationships directly, while linking to other data in other sources. • A federated model requires a good data management model where data from multiple sources can be integrated. What type of data should be federated? • Federated data might be data that needs to be tracked, but not as often or as vigorously as a CI's key attributes (extended data). • Data referenced by a CI to provide additional content related to the functionality of the CI, but not part of the CI itself (related data). The challenges of integrating multiple repositories (for example, reconciling diverse and redundant data) remain. Approaches used to reconcile diverse and redundant data include application programming interfaces (APIs) and Extensible Markup Language (XML) to facilitate data integration.

Table 2. Key CMDB features and representative vendor offerings

	Representative vendor offerings*							
Key CMDB features	nLayers	Relicore	BMC	HP (Mercury)	Cendura	Tideway	CA	IBM/ Collation
Data reconciliation	•	•	•	•	•	•	•	•
Data integration	•	•	•	•	•	•		•
Change detection	•	•					•	
Scalability	•	•	•	•	•	•		•
Application discovery	•	•	•	•	•	•	•	•
Continuous monitoring	•	•	•					
Discovery technology	•	•	•	•		•	•	•

* Not an exhaustive list. Based on industry trends and BearingPoint experience. Does not represent level of capability. Each organization must define detailedCMDB requirements, prioritize the requirements, evaluate and short listvendors before selecting a best-fit tool. Start with the key capabilities and scope of the CMDB implementation to define and include CIs that will provide the most business value in the short term (0–4 months). Employ an iterative process to assess the maturity of the CMDB and enhance it in alignment with your specific organizational requirements.

From our experience we expect that most Global 2000 organizations will need to federate between three and seven data sources. Further, over the next three years, because at least 25 percent of those organizations will be implementing CMDB initiatives without proper planning and organizational alignment, grassroots initiatives will emerge in various departments, for example asset, data center, networking and service desk. This will create multiple CMDB silos.

From our experience we expect that most Global 2000 organizations will need to federate between three and seven data sources. Further, over the next three years, because at least 25 percent of those organizations will be implementing CMDB initiatives without proper planning and organizational alignment, grassroots initiatives will emerge in various departments, for example asset, data center, networking and service desk. This will create multiple CMDB silos.

Selecting a CMDB solution: Guidance for vendor and tool evaluation

Before selecting a CMDB vendor and tool, consider the following:

- Establish executive-level buy-in and funding within the organization. Lack of executive-level commitment often is a pitfall in implementing CMDB solutions.
- Analyze and define an end-to-end configuration management process, with consideration given to identifying specific interfaces with change management, release management and IT asset management.
- Define process goals, objectives, roles and responsibilities, governance structure, and metrics.
- Document organization-specific requirements for the CMDB based on the defined process, interfaces and organizational dependencies, for example, funding, resources, and the ability to operate and sustain the CMDB.
- Define IT services and related infrastructure components required to provision services. It is critical to understand IT services from a business partner as well as an IT perspective. For example, if a trader accesses a research report every morning at 7:00 a.m., IT would need to understand all of the related infrastructure and application components required to provide that service to the trader.
- Understand both the one-time and ongoing cost implications from a process, organizational and technology perspective.

In an organization that has espoused Design for Six Sigma (DFSS) as a federated model to introduce major transformation projects, these steps are usually undertaken as part of the "define" phase of DFSS requirements definition, during which key performance indicators (KPIs) are developed.[2]

The recent mergers and acquisitions of vendors offering CMDB solutions is an effort to reposition and strengthen CMDB toolsets. No one solution is the answer, and vendors are adding to their portfolios in order to offer a comprehensive solution.

Key vendors that are now actively developing, evolving and merging to provide useable and manageable functionality have appeared in the CMDB space over several years. Some key CMDB tool capabilities and the vendors currently emerging as solution providers are listed in Table 2.

Financial services organizations have large, complex IT organizations with multiple databases that will provide information feeds to and from the CMDB. Considering the operational capability of operating, maintaining and sustaining the CMDB will be important. In our experience, some organizations are forming "configuration management" groups within IT infrastructure and operations to own, manage and govern the CMDB. It is crucial to designate a configuration management process owner and CMDB custodians who provide access control to information that is stored and changed in the CMDB.

Some overarching process, governance and organizational considerations include:

- Defining end-to-end process governance and key performance indicators
- Designing the configuration management process to support handoffs from one process activity to another, for example, change control integrated with release management to update information in the CMDB
- Securing the budget for both organizational and operational support of the CMDB
- Defining process roles and responsibilities and mapping them to your organization's functional roles
- Balancing automated discovery with change control reviews and approvals

The configuration management process can be the first step in unifying decision-making processes across IT infrastructure and operations. It is critical to view the CMDB as a decision-making enabler (a process-driven approach) rather than merely something "to buy and implement" (a purely technology-driven implementation).

[2]See the Spring 2006 issue of the BearingPoint *Financial Services Technology Journal*, page 45, for additional discussion of DFSS.

IT service as an offering: The emergence of the IT service catalog[3]

Many organizations have determined the importance of creating service level agreements (SLAs) and operating level agreements (OLAs) and managing multiple vendor underpinning contracts (UCs) in order to set expectations between the IT service provider and the business partner who is receiving and paying for the services. However, SLAs are often developed in a silo with little to no consideration within a comprehensive service catalog.

Service catalogs have recently appeared in organizations as Web-based lists of service offerings available to the end user. These service offerings comprise many infrastructure components and support offerings. These are typically listed with a predefined cost for "basic" services, with additional cost estimates available for offerings considered to be "exceptions" to those listed in the service catalog. Often, there is one view of a service catalog aimed at the business consumer, but there also needs to be an internal, IT-facing service catalog that includes the OLAs and UCs required to provision the service expected by the business consumer. A well-defined communications plan targeted at both the business and internal IT staff supporting the services is essential to support understanding of and adherence to SLAs and OLAs.

The challenge lies in reconciling the multiple SLAs that may exist in the organization to the actual services being offered and provided to the business partner or end user. In order to reconcile what may be multiple documents, spreadsheets and databases and to understand how the service catalog, SLAs, OLAs and UCs need to connect, vendors such as HP, BMC, newScale and Infra offer software-based solutions aimed at taking the guesswork out of creating a service catalog. These toolkits often come with wizards that facilitate the creation of templates and communications and implementation plans.

Before selecting a tool for a service catalog, however, it is important to consider the following:

- Defining the service level management process, which identifies all of the necessary process inputs, outputs, metrics and reporting, as well as any process interdependencies required to support and sustain a service catalog and its associated SLAs, OLAs and UCs.
- Understanding the one-time and ongoing costs, as well as the effort and focus required to develop and maintain an effective service catalog.
- Verifying that there are defined and accepted roles for a service level manager(s) and vendor manager(s) in the organization. These roles are critical in order to sustain and manage expectations between internal and external business consumers and the IT service providers.
- Aligning business drivers with IT services and subsequent IT service provisioning requirements.

How the CMDB and the service catalog relate to each other

Both the CMDB and service catalog initiatives require executive sponsorship, clear definition of process metrics and benefits, definition of process adoption impediments, continued awareness[4], and a realistic, step-by-step plateau-based approach to process design, execution and ongoing service improvement.

Service catalog objectives

A service catalog is a compilation of IT service offerings that are documented for a business-facing audience. The service catalog can be considered one of the "tools" or outputs of the service level management process.

An IT service catalog provides a mechanism for business units to select and consume IT services and make informed decisions on IT service capability and availability. The catalog opens a dialog between IT and the business units, supporting a "business partnership" with IT.

The first critical step in architecting your service catalog is to define an IT service offering. An IT service offering could be defined as a set of facilities, IT and non-IT, sustained by the IT service provider that fulfills one or more needs of the customer, supports the customer's business objectives, and is perceived by the customer as a coherent whole.

CMDB objectives

The CMDB provides a tool for IT operations to control, identify and manage the IT infrastructure components. This helps provide IT with the agility to adapt and provide scalable technology to meet evolving business requirements.

The CMDB can be used as an internal catalog by IT operations to account for, control and define CI relationships. A significant benefit of implementing a CMDB is understanding these CI values and attributes to provide a comprehensive view of service-impacting components. Therefore, if the infrastructure components required are known and are related to the provisioning, maintenance and support of a service offering, planning for and mapping out the CMDB can begin.

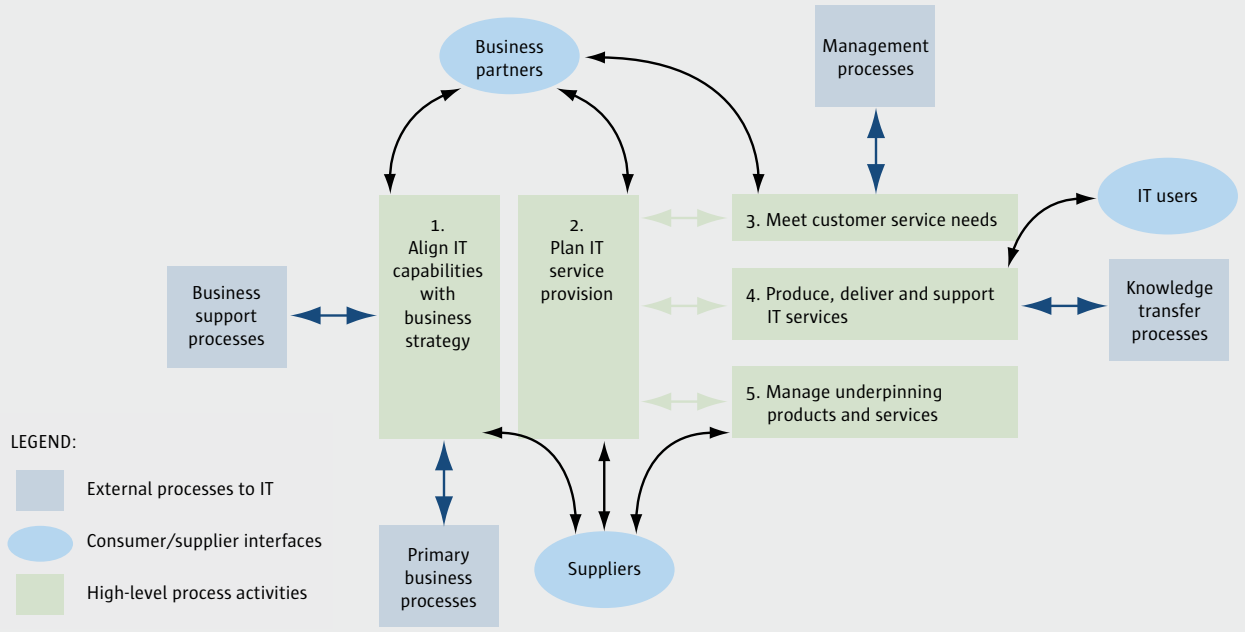
The IT services decision chart (Figure 2) depicts some key activities and interfaces that are relevant to both the service level management and configuration management processes. Each process and its related activities play a key role in the planning and delivery of IT service offerings and, subsequently, in the definition of a service catalog and the ongoing maintenance of a CMDB.

The IT services decision chart can serve as a tool to determine the order of implementation for the CMDB or service catalog. Decisions should be based on organizational, resource and budget constraints. For example, the first step should be to determine how IT

[3]In this context, a service offering is a comprehensive set of IT services available to the internal business consumer. See "Architecting the IT Organization" by M. Barbero, specifically the discussion about product management and catalogs, in the Spring 2006 issue of the *Financial Services Technology Journal*.

[4]Architecting the IT Organization" by M. Barbero, specifically section 8 "Using agile documentation and lightweight planning: The importance of the plateau definition," in the Spring 2006 issue of the *Financial Services Technology Journal*.

Figure 2. IT services decision chart



capabilities align with business strategy (number 1 in Figure 2) and what resources IT will require to provision the service (number 2 in Figure 2). If it is determined that the definition of a "service" is not available, then begin with defining the service offerings before designing a CMDB. This is an iterative process within which identification of the required IT service offerings and related infrastructure components should provide a gap analysis to determine the order of magnitude of the design and implementation activities required to implement either the CMDB or Service Catalog.

When setting priorities and selecting the order in which a CMDB or a service catalog should be implemented, it is important to view the decision-making process as an iterative cycle with a feedback loop at each stage of the decision cycle, allowing for continuous improvement and customization as each initiative progresses.

When setting priorities and selecting the order in which a CMDB or a service catalog should be implemented, it is important to view the decision-making process as an iterative cycle with a feedback loop at each stage of the decision cycle, allowing for continuous improvement and customization as each initiative progresses.

In our view, organizations are approaching the creation of a data repository for all the information that may be included in a CMDB as part of transformational change activities. These activities require data gathering, data cleansing, maintenance and ongoing management of a host of data elements required to gain insight into the infrastructure, both from an economic and management perspective. This is akin to setting up a bona fide "master data management" or MDM approach to the management of data pertaining to the IT environment.[5] Some organizations are taking the approach of setting up project work streams to design and build the desired data structures and database as part of a larger infrastructure transformation, with the goal of migrating this structure into a production environment where the database would then be utilized as the CMDB. ■

[5]Regarding master data management, listen to http://www.podtech.net/home/3480/master-data-management-turning-data-in-to-assets, a podcast from our very own Journal Editor, Alex Berson who with co-author Larry Dubov, also from BearingPoint, has published "Master Data Management and Customer Data Integration for a Global Enterprise," McGraw Hill, excerpted at http://searchdatamanagement.techtarget.com/generic/0,295582,sid91_gci1253049,00.html

Planning a pandemic response plan: Why everybody should have one

Abstract

The likelihood of a pandemic is small but not remote. A pandemic is a health emergency that has regional, national and, in extreme cases, global impact. Pandemics differ from other disasters in that they are typically longer, with the event and its impact lasting months or more. For industries such as insurance, banking and other financial services, it is essential to have careful and detailed plans for the necessary response and recovery of their operations should a pandemic event occur. This article provides insight into some of the essential considerations and overall pandemic response issues.

Planning is essential

A pandemic typically impacts personnel first—for example, people who are sick or cannot report to work because areas are quarantined. Systems will be unattended if the affected data centers are inside the quarantine area. Services cannot be restored when these systems fail. A key characteristic that sets a pandemic situation apart from other disasters is that pandemics are typically longer. Such an event and its impact can last months or more. This means that systems and services are not likely to be restored for quite some time.

For some industries, such as insurance, banking and other financial services, having careful and detailed plans for the necessary response and recovery of operations given a pandemic event is essential. This article provides insight into some of the considerations and overall pandemic response issues.

Likelihood of a pandemic

The likelihood of a pandemic is small but real, and given the relatively long time frame since the last pandemic outbreak about 90 years ago, the likelihood of an imminent one is high.

Today's large enterprises are heavily dependent on systems that execute millions of transactions. These systems require personnel to maintain and update them. A pandemic could cause devastating economic damages when the people who maintain the systems are not able to do so.

Pandemics are more prevalent than one might think. The well-known severe acute respiratory syndrome (SARS) epidemic originated in Guangdong Province, China, in November 2002 and quickly spread to other countries. In a few short months, SARS had spread to 26 countries. Quarantines were implemented in several provinces in China, Hong Kong, Taiwan and Singapore. The SARS epidemic lasted eight months.

Influenza pandemics have occurred every 20 to 30 years throughout history; the most serious were the "Spanish Flu" of 1918, the "Asian Flu" of 1957 and the "Hong Kong Flu" of 1968. Most recent and well-known is the "Avian Flu," commonly known as the "Bird Flu." Although Avian Flu has not impacted human life in an epidemic way, experts generally agree that a flu pandemic could occur in the next 20 to 30 years.

Pandemics can also be man-made. Given recent events, terrorism could be a significant potential cause for the next epidemic-like outbreak. While governments around the world are actively and diligently working to prevent such attacks from happening, it is prudent to plan for the possibility. A single successful terrorist attack could start a pandemic.

Nature of a pandemic

Experts including those at the World Health Organization (WHO) fear a worldwide outbreak of a pandemic, such as one resulting from a variant form of the Avian Flu or SARS. They are adamant that the question is not if but when this will happen.

Even if a vaccine is developed and disseminated, millions of people worldwide could die from such a pandemic. The effects on the economy and businesses could be catastrophic in the short term of approximately three to six months, depending upon the time frame of containment, vaccination and recovery. Psychological effects resulting from post-traumatic stress syndrome would likely persist for longer, even for years.

Today's large enterprises are heavily dependent on systems that execute millions of transactions. These systems require personnel to maintain and update them. A pandemic could cause devastating economic damages when the people who maintain the systems are not able to do so.

Effects on the economy and businesses

The effects on the economy and businesses are far reaching and include the following:

- Trading and transportation of commodities, food, products and people would slow due to sick or dead workers throughout worldwide trade chains, adversely affecting international trade.
- Supply chain management systems for just-in-time inventory would be adversely affected.
- Globalization factors would make the economic slowdown worse than at any other time in history when other pandemics have taken place.
- Medical treatment facilities, hospitals and clinics would be filled beyond capacity.
- Tourism and hospitality sectors would be in dire straits from lack of business.
- Schools and universities would close as students stayed in safer, contained environments or provided care for elderly family members and other affected people.

Financial sector impacts

Financial institutions would be affected in a variety of ways by a pandemic. Institutions would find it extremely difficult to deliver services due to absenteeism and massively volatile financial markets.

In addition, defaults on mortgages and corporate debt would rise dramatically. Worldwide liquidity would slow to a trickle as massive deflationary forces enveloped the globe. Credit risk premiums would rise sharply as credit demand plummeted.

Risk management challenges

All of these risks need to be managed by financial institutions. The types of risk associated with a pandemic are classified as operational risks, though these operational risks are so severe that they would also create market risks and credit risks for institutions.

Financial institutions need an overall integrated approach to pandemic risks, such as enterprise risk management, that addresses market, credit and operational risks.

Because of this, financial institutions need an overall integrated approach to pandemic risks, such as enterprise risk management, that addresses market, credit and operational risks.

Articles in the *Financial Service Technology Journal*, Volume 2, Issue 1, address some of these risk management challenges. In addition, another article in this current issue, "The new challenges of disaster recovery technology and their impacts on computing" by Daryouche Behboudi, discusses some of the many challenges and technology tools and controls that can be employed to address the pandemic scenario. These disaster recovery tools and controls focus on the IT part of business continuity planning within a pandemic response plan (also see the "Pandemic response plan" section below).

Regulations and checklists

As part of enterprise risk management, a number of regulations have been formulated to help financial institutions in the event of a pandemic.

The "National Strategy To Safeguard Against The Danger Of Pandemic Influenza"[1] developed by the U.S. Department of Homeland Security guides preparedness and response to an influenza pandemic, with the intent of:

1. Stopping, slowing or otherwise limiting the spread of a pandemic to the United States
2. Limiting the domestic spread of a pandemic, and mitigating disease, suffering and death
3. Sustaining infrastructure and mitigating impact on the economy and the functioning of society

Associated with this strategy, the U.S. Department of Health and Human Services has developed the Pandemic Influenza Planning Checklist. The basics of this checklist can form the basis of a pandemic response plan.

Pandemic response plan

The following summarizes the Pandemic Influenza Planning Checklist:

1. Plan for the impact of a pandemic on your business
2. Plan for the impact on your employees and customers
3. Establish policies to be implemented during a pandemic
4. Allocate resources to protect your employees and customers during a pandemic
5. Communicate with and educate your employees
6. Coordinate with external organizations and help your community

Proper pandemic planning requires the development of a response plan that incorporates elements of the above in detail for any given organization. The key to an effective pandemic response plan is to consider operations (at least core operations) and plan how they can resume in other regional locations or even other countries. Potential details include:

- Leadership succession plans may have to be developed in the event that leadership is incapacitated. The succession plan may be in the form of power of attorney or other legal authorization for emergency executive power.

[1]See http://www.whitehouse.gov/homeland/pandemic-influenza.html

- The insurance industry may need to incorporate plans for coordination with local, state and federal governments so that services to the insured, such as claims and medical procedure authorizations, can be provided.
- The banking industry will need to prepare for an unusually large number and amount of withdrawals and may need to provide an alternative way of delivering services to customers.

Two of the key components of an effective pandemic response plan are a solid communication plan and a pandemic alternate site (PAS). In emergency situations, employees often cannot communicate with each other, and actions are taken, or not taken, that cause damage to the enterprise. During the SARS epidemic in China, many companies suffered economic losses because their main site of operations was within the quarantine zone.

Communication plan in a pandemic situation

The communication plan in a pandemic situation is critical for directing operations. For smaller companies, a simple telephone directory card may be sufficient. However, for large and global enterprises, a voice bulle0 0. n board may be required.

Unless the enterprise has a data center that is outside of the quarantine zone, electronic communication channels may be interrupted because the systems are left unattended. The communication plan should have different tiers for information segregation. Security is critical in pandemic situations. Different levels of personnel may need to access different emergency communication sites.

Enterprises should examine their current operations, identify the critical components and determine how they can be resumed if their main data center is inside the quarantine area.

Keep in mind that the quarantine area usually starts at the local level and has the potential of rapidly expanding to the regional level. Therefore, a backup data center that is 150 miles away may soon be within the expanded quarantine area. The quarantine for the SARS epidemic in China was expanded quickly from Beijing to include several neighboring provinces. Hong Kong, which is more than 1,000 miles away, was quarantined within one week of the outbreak.

Pandemic alternate site (PAS)

Enterprises should rethink their disaster recovery strategy to incorporate requirements for pandemics. Companies that have a single global data center with a backup data center located just a few hundred miles away are particularly vulnerable. This is especially true in today's climate of consolidation.

The PAS is more than just a recovery site for systems. It should also have the capability for operations recovery. Unless every employee is issued a home computer, there is no guarantee that an employee can access company systems remotely. Even if all employees have their own home computer, there is no guarantee that they have the necessary software to operate effectively. After the SARS epidemic, a large global consumer electronics company in Korea changed all desktop computers to laptops so that, in case of a pandemic, the employees could work from home until further notice.

Enterprises should rethink their disaster recovery strategy to incorporate requirements for pandemics. Companies that have a single global data center with a backup data center located just a few hundred miles away are particularly vulnerable. This is especially true in today's climate of consolidation.

Finally, enterprises should also re-examine their storage strategy. Almost all companies have some critical data that is stored locally at the user's desktop in the form of Microsoft® Office Word or Excel files. This critical data will not be accessible if the office is inside the quarantine area. Many companies have invested millions in centralizing all storage and still find themselves not capturing some critical data. Switching to laptop computers only may be a quick and more economical solution.

Conclusion: Planning a priority for financial services

When planning overall business continuity and disaster recovery, a pandemic response plan should be an integrated part of this planning and associated testing. The potential for a serious and disastrous pandemic is real and could occur on a global scale. Because the financial services industry and financial services providers are the lifeblood of the global economy, pandemic response planning needs to be addressed as a priority within any business continuity and disaster recovery and preparedness effort. The checklists, regulations, guidelines and technology-related tools outlined in this article can help in such an effort. ■

How it (should) all start

Data center capacity planning: An introduction and brief survey of optimization techniques

Abstract

The activities of the financial services industry have led to rapidly increasing demands for more computing and storage resources at data centers. This arti-cle discuss how the industry is re-engineering data centers to support this trend and how new business models are developing to measure the activity at data centers. We introduce the concept of data center capacity metrics (DCCM) to explain how companies assess management of resources and how they should be adapted to support new demand. DCCM have to cover much more than rack density and power per-square footage. Other DCCM are provided along several different dimensions to provide practical guidelines to build efficient data centers. This article also describes other key factors in reducing unused infrastructure and overall data center operating costs. Finally, we discuss a capacity planning tool that was developed to model DCCM for planning execution purposes.

Escalating demands for IT

Business demands for IT are continuously increasing with a corresponding escalation of requirements for more computing, data communications, storage and server capacity. This trend has forced financial services companies to expand their data centers significantly at a pace unseen in previous decades. As a result, data center costs represent an increasingly larger portion of the IT budget. In fact, our data shows that more than 60 percent of the IT budget typically goes toward core data center costs, such as infrastructure, system and networking costs.

This said, companies have historically tended to build data center infrastructure in a rather fragmented manner. Most financial services companies have gone through several mergers and acquisitions that include a number of data centers with infrastructure that could have been migrated or decommissioned. Moreover, different infrastructure components are traditionally managed by different groups, leading to disproportionate build-out of each component (e.g., more network ports than needed to accommodate servers, resulting in more space and power to support idle network devices).

To a large degree, rationalization of data center capacity is therefore a matter of discipline in terms of retiring obsolete infrastructure. This being said, a number of financial services companies have gone down that path and now have to streamline computing, storage and communication capacity systematically in the data center after eliminating the waste of low-hanging fruits.

One of the primary cost drivers is related to power consumption, which typically represents more than 40 percent of operational costs in data centers. The proportion is expected to increase due to continuing increases in the cost of electricity.

Cost is not the only issue associated to increased power consumption. Heat dissipation has also become a key issue as the load on cooling systems has been consistently growing for the last decade. Until 1999, densely packed cabinets produced only about 2–3 kilowatts (kW) of heat. Then, the emergence of 1U servers made cabinets produce up to 8 kW of heat. Starting in 2001, the introduction of blade server technology led to a potential heat dissipation of 14 kW per cabinet (where they are densely packed). In the meantime, the upper limits of traditional cooling solutions have not improved much. They are currently about 3.6 kW per cabinet, although new cooling technologies are becoming more prominent.

Altogether, the cost of power consumption for computing devices and cooling systems accounts for 70 percent or more of the total cost of construction in a data center, whereas it only accounts for 15 percent in office buildings.[1] Unsurprisingly, as companies increasingly experience capacity constraints associated with insufficient power and cooling, the hosting service market has started shifting its traditional business model from a per-square-footage charging scheme to a per-kW one.

Business demands for IT are continuously increasing with a corresponding escalation of requirements for more computing, data communications, storage and server capacity. This trend has forced financial services companies to expand their data centers significantly at a pace unseen in previous decades.

[1]Gartner Data Center Facilities Cost Survey: 2006 Update.

In response to this trend and to alleviate capacity constraints, large financial services companies have started adopting innovative technologies and solutions, including:

- High density servers
- Server consolidation
- Server virtualization
- Advanced cooling systems (e.g., water-cooled racks)

Each of these remediation approaches has trade-offs. Selecting the right mix of solutions amounts to balancing constraints in the three dimensions of the problem space: space, power and cooling.

The challenge is to model how these factors can be managed to influence the overall data center capacity. The next section of this article presents the typical metrics and techniques that should be considered in defining data center capacity and efficiency.

The optimization program

A first (and naïve) approach for optimizing data center capacity is to express it as a nonlinear but positively correlated function of the three dimensions (space, power and cooling) of the problem space. This approach is incomplete because "space" not only is a function of the footprint used by equipment, but also encompasses other factors, including cabinet and rack density. So the problem space is really five-dimensional:

- Footprint (square footage)
- Cabinet density (cabinet/square footage)
- Rack density (rack square footage)
- Power (total kW consumption)
- Cooling (total BTU required)

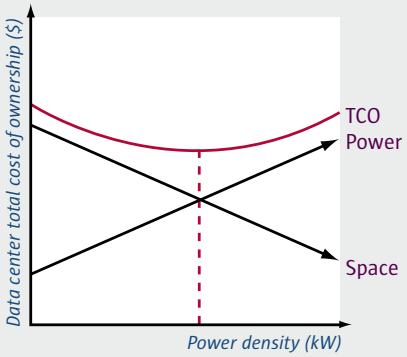
The optimization program is then about reducing the footprint, power, cooling product under the conditions existing in the data centers and the interrelationships that exist among the five dimensions, for example:

- Maximum usage for the computer hardware footprint (including servers, rack-mounted equipment, telecommunications frames) typically ranges between 25 percent and 30 percent of the total data center square footage.
- Densification is a limited solution to respond to space constraints. When the 6 kW threshold is exceeded, the total cost of ownership of data center services typically rises again and other solutions should be considered.

The relationships among the metrics are as follows:

- Typical power/sq ft relationship = 75 to 100 w/sq ft
- Typical cooling/space relationship = must match power output

Figure 1. Data center total cost of ownership correlated to a standard-size cabinet power density



Ideal balance between space utilization and power consumption estimated at about 6kW per cabinet

Source: BearingPoint research

- Typical rack unit/cabinet relationship = 42 (if cabinet is fully utilized, to be downgraded)
- Typical sq ft/cabinets relationship = 30 sq ft/cabinet (= 6 sq ft * 5 – infrastructure efficiency ratio)

The optimization approach can be conducted iteratively by performing the following three steps:

1) Divide data centers into homogeneous zones.[2]
2) Forecast change in resource requirements for each different zone of the data center as a result of organic growth, migration activities and new projects.
3) Use capacity management techniques similar to the ones prescribed in the IT Infrastructure Library (ITIL®)[3] for "capacity management" applied to data processing and storage (as described in the next section).

Note that this exercise in capacity planning will facilitate data center consolidation activities from exiting locations to key receiving sites.

Typical capacity optimization techniques

Inefficiency is a function of multiple factors, including overbuilt networks, excess storage capacity, white space in exit sites and unused capacity in new sites. Specific actions that companies can take to begin the process of improving the efficiency include:

- Sizing network ports according to required storage systems, servers and their standard network interface card (NIC) configurations in order to build network equipment and achieve the optimal efficiency ratio. Representative port densities for server, mid-range and mainframe platforms are:

[2]A zone is an area of the data center that is built to house equipment sharing a common set of power, space and cooling requirements. An example is a zone for high-density devices that have higher power consumption relative to their space requirements.
[3]The IT Infrastructure Library (ITIL®) is a customizable framework of leading practices with common definitions and terminology for the management of IT services. For further information, please refer to http://www.itil.co.uk.

  - Server: standard 2 ports, maximum 6 ports, average 3 ports.
  - Midrange: maximum 10 ports, average 5–7 ports.
  - Mainframe: standard 16 ports, environmentally hardened modules (e.g., Open System Adapter) on 30 percent of mainframe devices.

- Provisioning network bandwidth according to immediate and future demands in a data center. Data center consolidation will lead to more cost-effective provisioning of network bandwidth by combining a number of circuits that exist in multiple locations into higher bandwidth pipes connected to a small number of strategic data centers.
- Identifying the server utilization level to find opportunities to reduce server counts.
- Evaluating remaining equipment asset values to determine the potential for redeployment.
- Establishing standard cabinet loads. The standards should cover maximum power draw, weight, heat dissipation and rack units. Several organizations that have implemented these types of standards have been able to predict growth requirements more accurately, prevent hot spots and limit exceptions.
- Determining appropriate storage capacity management for optimal use of storage resources (configured and utilized), in particular to identify if existing storage systems can be migrated onto less equipment with more capacity.[4]
- Conducting careful migration planning to reduce the remaining white space in existing data centers or from moving into new data centers.

New technologies have been emerging over the last couple of years to support better utilization of core data center computing resources, including server and storage virtualization and utility computing.[5]

A capacity planning tool

BearingPoint has developed a data center capacity planning tool to forecast data center utilization. The tool can be configured for the different characteristics of the multiple data centers and homogeneous zones to be managed.

Modeling and optimization through scenario assessment and iterative heuristics can be performed in a matter of hours or days, depending on the availability of the information within the client organization. The tool provides the ability simulate capacity consumption at different points in time by adding servers, cabinets and square footage along with power consumption and cooling requirements. Capacity can be added by simulating the extension of zones, as well as of power and cooling capacity additions. The tool takes into consideration the following factors to provide a month-to-month forecast:

Metrics:

- Number of cabinets (i.e., the number of standard cabinets in the data center).
- Maximum possible number of cabinets, given the site and zone space constraints.
- Threshold: 80 percent of the limiting maximum possible number of cabinets.

Data entry points:

- Three data entry points feed the utilization forecast and result in data center resources consumption: organic growth rates, new projects and migrations.
- The two data entry points of space and power feed the data center resources forecast.

Characteristics for each zone:

- Servers/cabinet.
- kW/cabinet.
- Annual growth rate.
- Number of cabinets for the zone.
- Cabinet footprint (raw space in square feet), calculated based on the number of cabinets in the zone.

The capacity planning tool presents a graphic description of how any of the key metrics evolve over time (Figure 2). The logic relies on the homogeneity of each zone modeled, and thus capacity can be seen in terms of equivalent number of square feet, servers or standard cabinets, all constrained by the limiting factor for the zone, power or space.

The tool consists of an advanced Microsoft® Office Excel spreadsheet with macros that are configured to reflect the characteristics of the data center environment once it has been modeled into zones. The tool then needs to be baselined with the existing state of the data center (i.e., how much capacity is available and how much is used). Once these steps have been performed, the capacity planner can schedule capacity con-sumption over time and plan for capacity increases (e.g., power capacity addition or zone extensions). The utilization of resources is then graphically presented for each zone.

Typical time frames for the different implementation steps of the capacity planning tool include:

- **Configuration:** several days based on the complexity and size of the environment (e.g., the number of zones that are needed to model a site) and the availability of appropriate documentation (e.g., raised floor space plans).
- **Baselining:** several days based on the availability of utilization data for each zone. This step may require site visits if required data is not available.
- **Training:** approximately a day to understand the logic and implementation and maintenance steps.
- **Updating:** a few hours periodically (e.g., biweekly), dependent upon the frequency of changes in the environment.

[4]Storage capacity management, which should be examined in detail, is further discussed in the article, "Managing storage capacity: Tools and techniques," in this issue of the BearingPoint *Financial Services Technology Journal*.
[5]See the article, "Utility computing: Keys to successful implementation" in this issue of the BearingPoint *Financial Services Technology Journal* for more details.

- **Reporting:** a few minutes. The graphs are preformatted and can be easily extracted to Microsoft® PowerPoint.

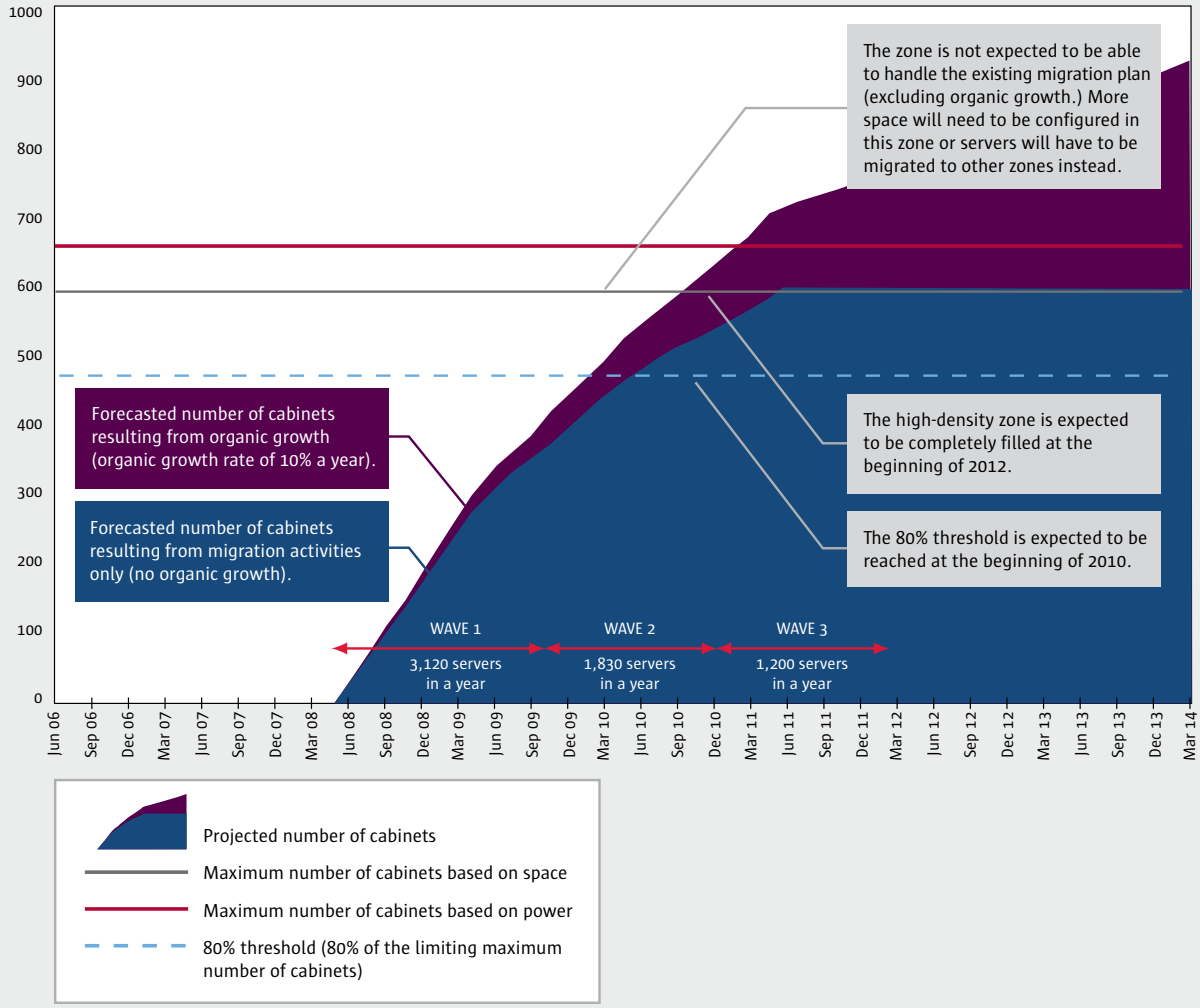
The tool has been successfully implemented at a large global diversified bank in order to forecast data center capacity utilization of more than 100,000 square feet of raised floor space split across four strategic sites of five zones each. Implementation required two weeks to model the zones properly and another two weeks of customization and configuration. The tool has been particularly helpful to analyze the capacity impact of migration activities from older sites being exited to the four strategic sites.

Conclusion

As discussed in the article "Data center capacity planning: An introduction and brief survey of optimization techniques" in this issue of the *Financial Services Technology Journal,* financial services organizations are now in a position to consider emerging new technologies to reduce the footprint and power consumption of data centers while meeting business demands in a more productive manner. They are introducing virtualization and utility computing technologies as they undertake the data center migration and consolidation efforts.

The current trend of increasing demands for IT will not slow down in the foreseeable future. IT management has the responsibility to review current data centers and identify how their configurations and capacity can be streamlined to reduce the run-rate while provisioning additional capacity with minimal capital spending. ■

Figure 2. Example of cabinet projections



The new challenges of disaster recovery technology and their impacts on IT infrastructure

Abstract

In today's financial services environment, the heightened reputation and operational risk of not having a com-prehensive and tested disaster recovery (DR) plan is simply too great to ignore. The dated model of relying on third parties is no longer viable when business requires recovery times of less than a day.

This article discusses various technologies and approaches that help the IT department develop and implement an effective DR strategy by looking at how that strategy impacts the enterprise IT infrastructure, particularly at the data center level.

Post-September 11, 2001, U.S. financial regulatory bodies—including the Federal Reserve, Securities and Exchange Commission (SEC), the Office of the Comptroller of the Currency (OCC) and others—renewed their emphasis on requiring financial institutions to have robust and actionable business continuity and DR plans.

As stated in the Federal Financial Institutions Examination Council (FFIEC) Business Continuity Planning booklet:

"Operating disruptions can occur with or without warning, and the results may be predictable or unknown. Because financial institutions play a crucial role in the United States economy, it is important their business operations are resilient and the effects of disruptions in service are minimized in order to maintain public trust and confidence in our financial system. Effective business continuity planning establishes the basis for financial institutions to maintain and recover business processes when operations have been disrupted unexpectedly.

Business continuity planning is the process whereby financial institutions ensure the maintenance or recovery of operations, including services to customers, when confronted with adverse events such as natural disasters,[1] technological failures, human error, or terrorism. The objectives of a business continuity plan (BCP) are to minimize financial loss to the institution; continue to serve customers and financial market participants; and mitigate the negative effects disruptions can have on an institution's strategic plans, reputation, operations, liquidity, credit quality, market position, and ability to remain in compliance with applicable laws and regulations. Changing business processes (internally to the institution and externally among interdependent financial services companies) and new threat scenarios require financial institutions to maintain updated and viable BCPs.[2]"

Disaster recovery (DR) refers to the technological component of a business continuity plan. It encompasses how IT systems are recovered in the event of a disruption.

This article discusses the challenges relating to DR technology in the data center and their impact on a financial institution's IT infrastructure. The article expands on a previous article published in the BearingPoint *Financial Services Technology Journal*.[3]

Elements of a DR vision

First and foremost, DR strategy must protect a firm's brand image and mitigate the risk of reputation damage. It should help protect the most important operations of the firm, allowing it to continue providing vital services to customers in the event of material business disruption.

- Build DR into everything the firm does. Consider new skill sets for some functional units to accommodate the long-term strategy, because every functional unit will eventually need to adapt; recognize that DR can become just an attribute within the firm's application and infrastructure architecture.

Business continuity planning is the process whereby financial institutions ensure the maintenance or recovery of operations, including services to customers, when confronted with adverse events such as natural disasters, technological failures, human error, or terrorism.

[1]See the article, "Planning a pandemic response plan: Why everybody should have one" in this issue of the BearingPoint *Financial Services Technology Journal.*
[2]The FFIEC Business Continuity Planning booklet is available at http://www.ffiec.gov/ffiecinfobase/html_pages/it_01.html#bcp.
[3]See "Designing and implementing a continuity of operations solution for financial services firms," *Financial Services Technology Journal*, Issue 3.

- **Consider DR as part of the organizational overall survival plan, not just a way to meet audit requirements.** Incorporate technology recovery, process survivability and human capital continuity as part of the overall IT business continuity plan. DR strategy is about the ability to manage the business during a disaster and the capability to survive a disaster event, not just recovering the business following an event. This proactive approach and mindset should allow the firm to build a viable program that meets all audit and regulatory requirements as a consequence of the program, as opposed to merely using a reactive approach to meet an audit demand.

- **Make sure the DR vision is scalable and compatible with the firm's technology standards.** Providing secure, flexible and scalable architecture to accommodate emerging technology is not just good practice to improve the enterprise's strategic positioning, it also helps introduce the kind of discipline that makes DR plans easier to implement.

- **Deploy using a plateau[4] approach for implementation.** Allowing for an adaptive deployment approach based on priorities (previous point) can secure support from the organization by showing a realistic approach to deployment of the DR plan.

- **Demonstrate cost-effectiveness.** Reusing technology assets where possible can help spell out the long-term opportunities to drive costs down.

- **Test, test and test.** Defining and implementing a realistic testing plan is a key to maintaining the viability of the DR strategy. Repeatable processes and procedures should be defined and published along with a well-defined reporting structure.

- **Pay attention to existing frameworks.** The enterprise must have robust service assurance and management processes and frameworks. The IT Infrastructure Library (ITIL[5]) or Control Objectives for Information and Related Technology (COBIT[6]) can guide an organization in improving and enhancing its service management processes.

- **Prioritize.** Providing tiered service levels based on a differentiated service model accommodates the needs of the business and protect vital assets of the firm. The following section expands on this point.

Tiered structure for DR

Since a firm's application portfolio is varied and includes critical and noncritical applications, a tiered DR structure based on an application's business impact analysis (BIA) ranking will result. Table 1 provides an overall framework that helps define six level of prioritization depending on an application's recovery time objective (RTO) and recovery point objective (RPO). (See "Disaster recovery terminology" for definitions.)

From vision to architecture — Data center and disaster recovery

The previous guiding principles are used to formulate a DR vision. They become the foundational requirements for the data center's DR conceptual architecture:

While the firm's DR vision will focus on survivability, using a multiple data center strategy will avoid unnecessary spending to meet these goals. Assets will be reused where possible, and costs will be driven down over a period of time by implementing cost savings and avoidance initiatives in conjunction with the DR program implementation.

The goal is to make DR an attribute of the firm's application and infrastructure reference architectures by considering it in all activities. A viable solution can not be created by one functional unit. Everyone in the organization must be made aware of the goals and

Table 1. Sample DR tiered structure

Name	RTO (hours)	RPO (hours)	Comments
Near real-time	<0.5	<0.5	Applications are stateless and latency tolerant; they do not cache data and are not mainframe-based.
Mission-critical	<12	<1	Application data is stored on replicated storage; replication of data is asynchronous; dedicated hardware is available for the application; the server image is stored on the storage area network (SAN), allowing the servers to boot from the SAN.
Business-critical	<24	<24	Application data is stored on replicated storage; replication of data is asynchronous; dedicated hardware is available for the application; the server image is stored on the SAN, allowing the servers to boot from the SAN.
Business-required	<48	<24	Application data is stored on replicated storage; replication of data is asynchronous; dedicated hardware is available for the application; the server image is stored on the SAN, allowing the servers to boot from the SAN.
Standard	< 72	< 24	Application data is stored on a tape management system or proprietary tape (e.g., AS/400, Tandem) or better (SAN-replicated); servers are non-dedicated (re-purposed non-production servers) or vendor-provided.
Best effort	> 72	> 24	Application data is stored on a tape management system or proprietary tape (e.g., AS/400, Tandem); servers are drop-shipped as needed.

[4]See "Architecting the IT organization" by M. Barbero in the Spring 2006 issue of the *Financial Services Technology Journal*.
[5]See http://www.itil.co.uk
[6]See for example: http://www.isaca.org/Template.cfm?Section=COBIT6&Template=/TaggedPage/TaggedPageDisplay.cfm&TPLID=55&ContentID=7981

encouraged to understand and take responsibility for making the firm's DR program a highly viable and cost-effective solution.

Overall, in designing the DC architecture, a mature DR strategy for a financial services firm will have to consider the following views[7]:

- **Facilities.** A three-site model that should include a primary site, a satellite site and a recovery site is the most robust. Hardware at the satellite site is dedicated and running in an active-active model with the primary site for high availability. Sufficient space is provisioned at the recovery site for recovery of all systems. All data centers are designed so that they qualify for Tier 4 data center classification.
- **Data network[8].** The network environment linking DCs should be fully redundant and resilient with no single point of failure. Bandwidth is more than sufficient for current and future needs. Real-time monitoring, alert notification and escalation are automated. In addition, data centers will require large redundant bandwidth that can be logically partitioned to separate Internet Protocol (IP), voice and replication traffic. Buying managed, fully protected synchronous optical network (SONET) services is the most economical way to achieve this. At minimum, an OC-48 must be provisioned. The bandwidth can be logically partitioned and each logical partition can be assigned to carry specific traffic, such as IP or storage. Branch to data center connectivity also needs to be considered. Frame relay or dedicated time division multiplexing (TDM) circuits will not be economical when connecting to multiple data centers. Only multiprotocol label switching (MPLS), with its ability to provide any-to-any connectivity, will be cost-efficient and allow for automatic failover to the DR site.
- **Computing systems.** A consolidated and virtualized computing environment should be provisioned for most applications, providing scalability and flexibility with reduced time to market for server setup and streamlined maintenance; moves, adds, changes and deletes (MACDs); and DR processes.

Disaster recovery terminology

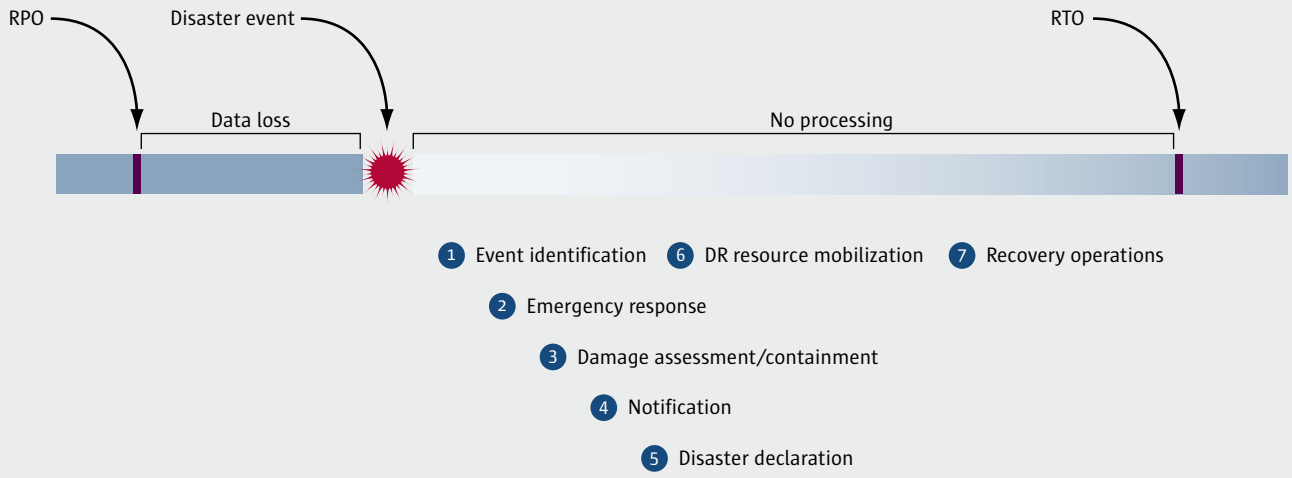
Recoverability — The degree or extent to which a service can be restored to an operational condition after a planned or unplanned interruption.

Recovery time objective (RTO) — A measurement of the maximum elapsed time between the occurrence of a disaster and when the affected processes are restored. For example, a service with an RTO of eight hours experiences a failure at 10:45 a.m. Functionality of the service must be restored to the customer prior to 6:45 p.m.

Recovery point objective (RPO) — A measurement of the maximum elapsed time of data loss resulting from a disaster. For example, a service with an RPO of 24 hours experiences a failure at 11:35 p.m. on a Thursday. The data used to recover that service must be more recent than 11:35 p.m. on Wednesday.

Figure 1 schematically describes the concepts of RTO and RPO.

Figure 1. RTO and RPO



[7]For a definition of "View" see for example "Enterprise architecture notation — Let's get physical", *Financial Services Technology Journal*, Issue 4.
[8]For more on the various relevant data networking technologies mentioned in this section, see "New advances and techniques to strengthen network resiliency: A primer" in this issue of the BearingPoint *Financial Services Technology Journal.*
[9]See "Configuration management database: Latest developments and trends" in this issue of the BearingPoint *Financial Services Technology Journal*.

- **Storage.** Dedicated DR storage capability should be provisioned for all the production data at the DR site. Full replication and recovery capability with redundant network connectivity and bandwidth is required. Tape or virtual tape backup systems should be deployed in both production and DR sites. No single point of failure should exist at either of the data centers.

- **Voice services, help and service desks.** Help and service desks and network operations centers (NOCs) must be distributed among the various data centers and have their own DR plans so that they can function in the event of a disaster. A robust federated and distributed configuration management database (CMDB) will help keep the two data centers in synch or, if not, allow easy discovery and reconciliation of differences[9].

  A wide range of applications based on Extensible Markup Language (XML) provide enhanced contact center services. Soft phones increase accessibility, flexibility and scalability and provide the ability to work remotely.

- **DNS.** The domain name system (DNS) architecture must be engineered to allow for automatic failover and rerouting to the DR site when the primary data center is down. Appliances need to take over as primary DNS servers and automatically reroute the traffic by assigning the DNS name to the IP addresses at the DR site.

Finally, DR consideration in the DC architecture leads to assessing various hosting scenarios such as the ones presented in Table 2. Each scenario has advantages and drawbacks in terms of complexity, cost and the level of protection they provide. For example, a three-site model needs to be implemented that includes synchronous replication of the production data to a satellite site and asynchronous replication to a recovery site. In that configuration, recovery steps are highly automated, and the solution is tested at the process level and includes upstream and downstream systems.

To support an effective multiple data center strategy, floor management tools must be included in the implementation plan. The benefits of deploying such tools include:

Table 2. Hosting scenarios

Scenario	Description (hours)	Pros	Cons
Vendor recovery	The DR data center is located at a vendor site. In the event of a regional disaster, the service may not be available if premium levels of service were not negotiated.	• Lowest overall costs • Limited capital expenditure • No idle hardware	• 72-hour-plus recovery times • Limited ability to conduct tests • Hardware tends to lag and costs more to refresh • No guarantee of availability if other clients have already declared • Must leave vendor site 30 days after declaration
Internal DR site	Asynchronous one-way replication from production to DR site.	• Better recovery times, but not synchronous • Hardware can be multi-purposed • Greater control of the environment • Can enforce security requirements	• Disaster impacts all production • DR not running production environment under a load (could experience failures during recovery) • Costs more than scenario 1 • Idle hardware
Dual data centers	Asynchronous two-way replication between the two data centers.	• Can switch load between A and B based on maintenance and DR testing needs • Includes failover capability • Can declare partial disaster, impacting only half of production • Can validate/test solution frequently with little impact	• Greater network requirements • May require more hardware and storage • Costs more than scenarios 1 and 2 • Idle hardware • More complicated to support
Three Data Centers (optional for institutions requiring continuous operations, such as trade execution or market data operations or operating in an environment rated high for pandemic disruptions)	Asynchronous two-way replication between data centers A and B; synchronous replication between data centers A and C.	• Highest level of redundancy of any option • Near-zero recovery time to satellite site C • Hardware can be multi-purposed • Can switch load between A and B for non-synchronous systems • Can declare partial disaster	• Highest overall costs • Cannot switch workload for systems requiring synchronous replication • Idle hardware • More complicated to support

[10] See also "Data center capacity planning: An introduction and brief survey of optimization techniques" in this issue of the BearingPoint *Financial Services Technology Journal*.

- Improved reporting
- A visual tool to assist with configuration management
- Greater access to data center information from online capabilities

These tools are also useful to improve utilization of DCs and planning for site capacity[10].

Impact of DR on the rest of the IT infrastructure

Computing infrastructure

Architectural decisions regarding the computing infrastructure that must be part of developing a DR architecture include:

- **Migration to blade servers.** Although not strictly necessary, deployment of blade servers can result in significant savings (between $5,000 and $6,000 per server, including the cost of water-cooled racks). These savings will reduce the cost of building out a DR infrastructure.
- **Virtualization[11].** Virtualization is fast becoming a prerequisite of any DR deployment. Virtualization will result in:
  - Lower total cost of ownership for servers
  - Higher server utilization
  - Increased operational efficiency
  - Improved manageability
  - Faster recovery of virtual machines as opposed to traditional servers
  - Extension of virtual resource pools across the production data center and its high availability satellite site, resulting in automatic failover to the satellite site
  - Ability to reuse test and development machines for production by decoupling the operating system/software from the underlying hardware
- **Boot from the SAN.** The SAN can store server images for all test and production servers and enable booting servers from the stored image via Preboot Execution Environment (PXE). The benefits include:
  - Better image management
  - Ability to reuse test machines for DR
  - Enhanced security through implementation of critical patches
- **Automated provisioning, configuration and monitoring.** IT operations are facing the daunting operating challenge of managing multiple data centers with potentially thousands of servers that need to be provisioned, configured and monitored to make sure that they are, for example, at the same operating system patch level. Configuration drift has been the single cause of failure of DR infrastructures. However, tools such as BladeLogic Operations Manager to automate provisioning and configuration and Tripwire® Enterprise, to monitor servers continuously for configuration drift help IT operations get a handle on the problem. Benefits include:
  - Consistency between the production and the DR environment
  - Policy-based provisioning of the operating system and the full server stack
  - Quick identification and repair of misconfigured servers
  - Rollback of changes as appropriate
  - Tracking of configuration drift over time
  - Configuration updates via a sophisticated packaging technology that provides the ability to deploy software as well as selectively undo complex changes
  - Scanning and patching of servers across different operating systems

Storage

DR is not possible without replication of critical data. Real-time and mission-critical data must be synchronously replicated using such tools as EMC Symmetrix® Remote Data Facility (SRDF™) or IBM Metro Mirror. Both SRDF and Metro Mirror have distance limitations of around 60 miles. This distance, however, does not meet the DR guidelines of U.S. regulatory authorities, which generally require DR sites to be 100 miles or more from the primary production site. In these circumstances, asynchronous replication using EMC SRDF-A or IBM Global Mirror must be used.

One of the unfortunate results of using asynchronous replication is the requirement of using a minimum of two times the required storage at the DR site. Due to the time lag in creating business continuity volumes (BCVs) at the DR site, at least two copies of the BCV (the current and the prior volumes) need to be maintained at all times.

To reduce costs, therefore, it is necessary to identify mission-critical data during the business impact analysis. Table 3 represents the type of replication necessary per DR tier.

Storage virtualization will also help to manage and control the cost of storage. Like server virtualization, storage virtualization decouples the data from the underlying physical hardware, potentially allowing for the use of more cost-efficient storage media. Other benefits include:

- Increased management efficiency

Table 3. Replication type per DR tier

Name	RTO (hours)	RPO (hours)	Storage type
Near real-time	<0.5	<0.5	Synchronous replication
Mission-critical	<12	<1	Asynchronous replication
Business-critical	<24	<24	Asynchronous replication
Business-required	<48	<24	Asynchronous replication
Standard	< 72	< 24	Tape
Best effort	< 72	< 24	Tape

[11]On virtualization and other related topics, see "Utility computing: Keys to successful implementation" in this issue of the BearingPoint *Financial Services Technology Journal.*

- Better capacity management by allowing applications to access storage capacity beyond their SAN "island"
- Applications insulated from physical changes to the storage infrastructure
- Avoiding single points of hardware failure
- A single interface for storage volume configuration, management and service tasks

The issues related to that matter are further developed elsewhere in this Journal[12].

Database recovery

Databases either on stand-alone servers (virtual or physical) or in clusters represent a DR challenge. Even mainframe-based databases will require some level of manual intervention to re-initialize the database and get it ready for use by the applications. Use of such tools as Oracle® High Availability (HA) extensions will make this task easier by automatically copying the transaction logs to the DR site. However, versions of Microsoft® Structured Query Language (SQL) Server™ prior to SQL Server 2005 lack this capability and require extensive manual intervention so that they can be readied for use. The impact of database recovery and the time for manual intervention must be included in the RTO and RPO calculations.

Data center operations

Data centers can be run either on an active-active or active-passive mode:

- Active-active data center pairs refer to the situation were both data centers are online and processing transactions.
- Active-passive data center pairs refer to the situation where only one data center is online and processing transactions.

Table 4 summarizes the pros and cons of active-active and active-passive DR architectures.

Due to the high cost and level of complexity, active-active data center configurations must be reserved exclusively for those real-time or mission-critical applications, such as trade clearance, where such investments are justified.

Conclusion

Of all the domains in which disaster recovery needs to be provided, financial services firms must especially protect themselves against a disastrous disruption of data centers. The potential risks of such a disaster to the firm's reputation and operations are simply too great.

Design patterns and technologies are now available to help the IT department meet today's new, more aggressive DR requirements. The key to an effective DR strategy and posture is to incorporate DR as an attribute of everything a firm does, not just as an afterthought. ■

Table 4. Pros and cons of DR architectures

	Active-active	Active-passive
Description	• Both data centers are online and processing transactions • Provides real-time RTO and RPO	• Only one data center is completely online at any time to process transactions • Provides near-time RTO and RPO
Pros	• Fast recovery and agility for business • No downtime and increased capacity • Systems always updated and in sync • Significantly decreases productivity loss, reduces revenue loss and improves financial performance • Complete transparency of any disaster or maintenance scenario	• Provides immediate step toward active-active solution • Less risk and complexity than active-active • Recovery processes will improve over time • Moderate cost to implement as compared to active-active • Lower monitoring cost than active-active
Cons	• Extremely diligent change management process and policy required • Higher risk of production-related issues due to multiple tiers • High cost and high complexity • Requires extremely complex monitoring, policies, procedures and trained staff • Must keep systems < 40% utilization, resulting in reduced efficiency • Additional resources required for testing and maintenance • May require some application recoding • Technology is not currently available or mature	• Extremely diligent change management process and policy required • Potential productivity loss, revenue loss and impact to financial performance during the recovery period • Additional resources required for testing and maintenance • Longer RTO and RPO than active-active

[12] See "Managing storage capacity: Tools and techniques" and "Data center capacity planning: An introduction and brief survey of optimization techniques" in this issue of the BearingPoint *Financial Services Technology Journal.*

Data for decision

Scenario analysis for decision making: Outsourcing versus cost takeout

Abstract

Cost reduction efforts usually occur in a tactical, often annual cycle, while outsourcing initiatives happen over a longer, multiyear time frame. In this article we use the real-life example of a leading financial services firm to illustrate how conducting a cost-savings investigation doesn't preclude assessing and taking advantage of outsourcing options.

We examine how large financial services IT organizations can compare cost reduction initiatives and outsourcing scenarios, particularly in situations where no "easy" savings can be achieved, because obvious inefficiency or redundancies are lacking. This article first details and illustrates how to set up the financial model and develop a number of scenarios for comparison, then discusses the benefits and the limits of such models.

Identifying cost takeout opportunities

Consider the IT infrastructure organization in a leading financial services firm. This organization offered none of the obvious opportunities for cost reduction: no significant merger-related redundancies, no geographic overlap, no gaps in inventories or vendor management, and no proliferation of overlapping projects. However, senior management still wanted to explore cost reduction initiatives, including those beyond the scope of traditional cost takeout approaches.

The first step in identifying cost-savings opportunities to explore was a review of the current state of the organization. The data collection included financial information for each IT infrastructure domain. In our example, each domain provided data about its organization, operational procedures, finances and current significant projects.

Thorough data collection verified the integrity of this approach and produced both the financial and technical/organizational baselines. The financial baseline included headcount and line items considered in the model, and the technical/organizational baseline helped identify cost takeout opportunities within the company.

Competitive review of a number of the IT organization's peers completed the internal documentation of the baseline. In our experience, such competitive analysis must be aimed at specific, targeted data points, rather than be conducted as an extensive benchmark study. This allows the organization to draw up a list of assumptions from comparable organizations and substantiates the initial quantification of the cost-savings target.

By the time data collection was completed, major cost reduction initiatives had been identified. These are usually a combination of vendor renegotiations, control of high-profile expense items (for example, cell phones) and simplification of the technology architecture. At this point, the proposed initiatives were discussed and validated with the impacted IT domain leads. A smaller set of initiatives were selected for "deep-dive" analysis to validate assumptions made during the competitive analysis. For each initiative, a baseline, a cost reduction target and an implementation timeline were developed for preparation of the financial model.

In a highly inefficient organization, or one that has undergone several significant organizational changes without much concern for cost control, it is possible to

In a highly inefficient organization, or one that has been subject to several significant organizational changes without much concern for cost control, it is possible to achieve cost reductions of 20–30 percent in the baseline over the previous year.

achieve cost reductions of 20–30 percent over the previous year's baseline. In our example—an organization that has been kept under tight financial management—it is typically unlikely to reach even 6–8 percent baseline cost reductions. Accounting for resistance from middle managers, the cost-saving target can be expected to drop below 3 percent, often reaching a threshold under which further investigation is pointless.

However, including outsourcing as an initiative breathes life into the cost reduction effort. The objective is to explore outsourcing scenarios that can help reduce the baseline below what conventional cost reduction efforts can yield.

Building the outsourcing scenarios

The approach boils down to mapping cash flows quarter by quarter over sufficient time to show the outsourcing impact in steady state. The current example uses a five-year model.

Initial investments required to realize the cost reduction initiatives are spread over a time period consistent with implementation realities. In the same fashion, the savings ramp up progressively over time, preferably following a conservative agenda. The model is similar in spirit to mapping cash flows stemming from the outsourcing of one or several IT domains. The initial investment, however, and the implementation road map require significantly more detailed assumptions.

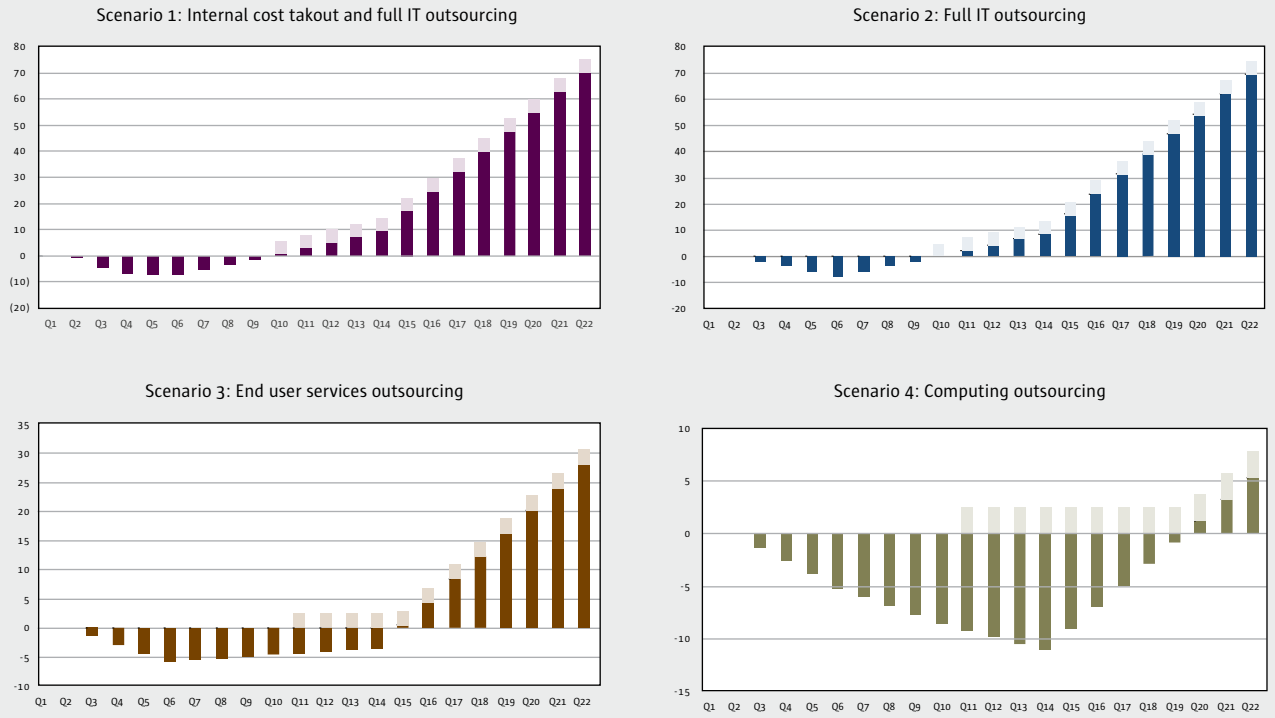
Investment costs are broken into phases: standard request for proposal (RFP), baselining (including service level agreements, transfer and transition, vendor transition, and transformation. Additional costs include contract termination and transfer costs, net equipment write-offs, legal costs, and a contingency budget.

Savings should materialize progressively, usually in the third year of the outsourcing process. For each IT domain, savings will vary depending on the estimated cost of service (for instance, using compensation information and service catalog data points) and typical outsourcer rates.

Once the model is built, it is possible to look at a combination of outsourcing the IT domains with an "internal" scenario, for example a standalone cost reduction scenario.

The charts in Figure 1 illustrate the profiles of some scenarios in this example. Scenario 1 represents full outsourcing of IT infrastructure and implementation of identified cost reduction initiatives. Scenarios 2, 3 and 4 are purely outsourcing scenarios (no cost reduction initiatives were implemented); within scope of the outsourcing, respectively, were the full IT infrastructure, all end-user services (workstation support and desk side support), and all computing services (mainframe and distributed systems).

Figure 1. Example of scenario models



Savings should materialize progressively, usually in the third year after the beginning of the outsourcing process. For each IT domain, the savings will be different depending on the estimated cost of service (for instance, using compensation information and service catalog data points) and the typical rates that outsourcers can provide.

Scenarios 3 and 4 show little value on a standalone basis because the first positive cash flows are too distant. However, scenario 2 can generate more immediate results. As such, it is worthwhile to more deeply investigate each scenario's feasibility by fine-tuning assumptions with the IT domain leader.

Conducting scenario analysis

The goal is to facilitate decision-making in terms of financial management. One possibility is a workshop session with IT infrastructure domain head to validate scenarios by ranking them against a set of objectives. A quality function deployment (QFD) tool can be used to facilitate this process (see "A practical QFD workshop").

Often, the final decision is to mix cost takeout initiatives with one of the outsourcing scenarios or to proceed only with cost takeout. Outsourcing represents a long-term effort for the company to streamline costs, while standalone cost reduction represents a number of short- to medium-term initiatives to keep costs under control. One is a long-term commitment with strategic implications; the other is a short-term tactical program.

The value of such scenarios is that they permit consensus between IT and senior management, reconciling operational constraints with financial management objectives:

- Agreement on the value of cost reduction initiatives—under the pressure of outsourcing, IT management can be expected to make significant efforts to demonstrate that an internal cost reduction program can yield satisfying return on investment.
- Agreement on the validity of cost reduction targets—such models "put everything on the table." Once all cost reduction possibilities have been explored and the only remaining possibility is outsourcing, it is unlikely that senior management will ask for unrealistic targets.

A practical QFD workshop

The defined scenarios in our example combine outsourcing and cost takeout efforts that resulted from analysis of the current state and creation of the financial models. In the quality function deployment (QFD) table (Figure 2), they appear in columns.

After presenting the models to the IT domain leads who participated in the workshop, a brief 20-minute brainstorming session identified the strategic objectives of implementing one of the scenarios. Most of the objectives were in line with the IT infrastructure group's overall strategy as defined at the firm level (for example, respect for SLAs and significant cost takeout).

The next step was to weigh each objective and report this weight in the column to the right of the QFD table. A higher weight indicates the objective is more important, a lower weight means it is less important. In our example, the IT domain leads each attributed weights from 1 to 5 to the objectives. The sum of all marks became the weight, with 43 for the "heaviest" objective (significant cost reductions) down to 1 for the "lightest."

The final step was for the IT domain leads to assess and agree on each scenario's overall impact on the objectives. For instance, does scenario 1 have a strong, medium or weak impact on "significant cost reductions"? To obtain discernable results, we selected 9 points for strong, 3 for medium and 1 for weak. The final decision is multiplied by the weight of the objective.

The QFD session showed that the "everything" scenario and the "internal cost takeout" scenario were the most valuable to the IT domain leads, thus validating the cost takeout efforts and opening the door to a deeper look at outsourcing.

Figure 2. Example of scenarios ranking

Standard 9-3-1		
Strong	■	9.0
Moderate	■	3.0
Weak	▼	1.0

OBJECTIVES			SCENARIOS						WEIGHT		
			1. Everything	6. Bear Stearns internal alternative	2. End-user computing and connectivity services	5. Field services and selected IT operations	3. Computing and core systems	4. Diverse point solutions	Weight	Rank	
			1	2	3	4	5	6	1	2	
	Mandatory	1									1
	Significant cost reduction ($10–25MM)	2	■	■	■	■	■	■	430	1	2
	Acceptable SLA performances	3	■	■	■	■	■	■	360	2	3
	Acceptable risks	4	■	■	■	■	■	■	150	3	4
	Enables and adapts to current and future business needs (including scalability for SLA)	5	■	■	■	■	■	■	50	4	5
	Significant cost control improvement	6	■	■	■	■	■	■	30	6	6
	Commercially available	7	■	■	■	■	■	■	10	10	7
	Allows future re-integration	8	■	■	■	■	■	■	10	10	8
	Optional	9									9
	Takes the people dimension into account	10	■	■	■	■	■	■	60	4	10
	"Promotes service culture"—enhances and enables servic quality and monitoring	11	■	■	■	■	■	■	30	6	11
	Promotes operational excellence (standards, single point of contact/accountability, etc.)	12	■	■	■	■	■	■	30	6	12
	Enables demand management	13	■	■	■	■	■	■	30	6	13
	Spread technology obsolescence risk	14	▼	■	▼	▼	■	■	10	10	14
	Road map to reqs	1	757	753	679	643	453	429			
	Rank	2	1	2	3	4	5	6	< OVERALL RANKING		
			1	2	3	4	5	6			

A close look at our example

A global investment bank engaged the authors to perform a cost reduction assessment and explore outsourcing opportunities for the IT infrastructure. The CFO and CIO had agreed to a target of a 10–15 percent reduction in the annual expense baseline. During the three previous years, however, the baseline had already been reduced by 10–20 percent, and the bank had just performed an inventory cleanup. In other words, there was not much else to be taken out.

Through a traditional cost reduction assessment we identified a series of opportunities that could yield 8–12 percent from the previous year ($7 to $11 million with an initial investment of $3 to $5 million). None of these opportunities were "low-hanging fruit" because they involved contract renegoti- ations, automation of a number of support activities and some outtasking.

The explored outsourcing scenarios, on the other hand, showed potential to create more value than the internal cost takeout scenario: a reduction of $30 million in the baseline in the fourth year for an initial investment of $30 million over the first three years.

In net present value (NPV), the scenario of complete outsourcing was valued at $100 million, while the internal cost takeout scenario was valued at $43 million. Management decided to perform only the internal cost takeout program since it required only a modest investment and did not carry any significant risk to the busi-ness. Although the cost reduction initiatives seemed difficult to execute at first, they turned out to be very achievable for the technology leads, as opposed to the outsourcing options.

The main benefit of modeling outsourcing options and cost reduction initiatives is gaining consensus on what cost management targets are reasonable and should be set as objectives for the IT infrastructure organization.

- Preparation for outsourcing — while the focus is on cost reduction, a secondary benefit is that the analysis provides a first look at outsourcing. As such, it gives senior and IT management a framework to structure their thinking.

Consensus on objectives

The model does not address the details that a true outsourcing feasibility assessment would require. As such, its audience should be limited to the most senior IT infrastructure circles, such as the CIO, the CTO, the heads of a technology domain and selected financial management personnel.

To further explore outsourcing as an option, the firm would need to get actual quotes from outsourcing service providers and evaluate the risks and costs for every position and process. This is a tedious effort that requires discrete compensation review to avoid alarming IT infrastructure personnel.

The main benefit of modeling outsourcing options and cost reduction initiatives is gaining consensus on what cost management targets are reasonable and should be set as objectives for the IT infrastructure organization. ■

Managing storage capacity: Tools and techniques

Abstract

Risk mitigation, business continuity requirements and digitization, as well as recordkeeping requirements, are driving the need for extensive storage space. In addition, much, if not all, business-critical and regulated data needs to be mirrored, replicated, backed up, archived and retained. These requirements call for even more disks and tapes.

Nevertheless, as a result of poor capacity management processes, improper tier allocation, confusion over storage metrics, uninformed purchase decisions and out-dated backup policies, a lot of storage space is wasted. This article offers seven techniques to assist in dealing with the exponential growth of raw data that most organizations are facing today, as well as an overview of a storage capacity planning tool successfully used in real-life situations.

Introduction

Storage demand continues to rise as risk mitigation, business continuity and recordkeeping requirements expand. To reap the benefits of modern storage technologies and avoid costly investments in unnecessary hardware, capacity usage should be tightly controlled. We have identified seven techniques for addressing data growth, along with a storage capacity planning tool.

Seven techniques to manage storage capacity efficiently

1. Control the provisioning and decommissioning processes for storage.

Common sense suggests that, before adding capacity, organizations should avoid wasting space and provisioning time by implementing appropriate controls and mapping capacity to retired servers systematically to put them back in the pool.

Yet our experience with clients shows that utilization of allocated capacity is often at low levels (below 50 percent) and the allocation process—when it exists—is rarely implemented correctly. In fact, while most organizations have established written policies and procedures governing storage capacity allocation, they generally fall short of reaching their objectives.

However, capacity provisioning and de-provisioning policies and procedures can be effective when the following key factors receive due consideration:

- Storage policies should be embedded with provisioning and decommissioning processes to verify that the appropriate activities are carried out, such as creating change records and updating the configuration management database (CMDB).
- The subgroup of the IT organization managing storage (the "storage group") should be involved early in IT projects so they have the time to appropriately configure storage infrastructure to avoid capacity waste.

When existing storage capacity is evaluated, the analysis almost always identifies capacity that has not been properly decommissioned. For example, servers are often decommissioned by other IT groups without the storage group being informed, resulting in capacity remaining allocated and locked but unused. To grant the storage group real authority so that storage policies can be more effectively enforced, consider these key factors:

- The storage group must play a role on architecture review and change advisory boards, have the appropriate sign-off authority, and be backed by appropriate governance and support from IT executives.

As a result of poor capacity management processes, improper tier allocation, confusion over storage metrics, uninformed purchase decisions and outdated backup policies, a lot of storage space is wasted.

When existing storage capacity is evaluated, the analysis almost always points to the existence of capacity that has not been properly decommissioned.

- Storage management tools must be leveraged to help identify configured yet unmapped capacity. This must be a recurring task for the storage group.
- Storage chargeback must be implemented to provide users with incentives to communicate decommissioning activities to the storage group.

2. Reduce the number of reporting metrics. We recommend focusing on file system capacity, raw disk capacity and capacity efficiency ratio.

Multiple storage capacity metrics available in capacity reporting tools can be confusing to non-experts. How can differences between allocated, configured, mapped, assignable, assigned, raw, primary, etc., be simply explained? And, in addition to reporting errors and inconsistencies from one reporting tool to the other, significant differences can exist in the definition of such metrics.

To avoid confusion, users should challenge reports generated by the different software tools and stick with the following metrics:

- File system capacity—the amount of space visible to the computer.
- Raw disk capacity—the amount of raw disk space needed to provision the file system capacity, including all the capacity used for disk protection and replication.
- Capacity efficiency ratio—the raw disk capacity to file system capacity ratio, reflecting the degree of redundancy in a particular configuration or as a whole.

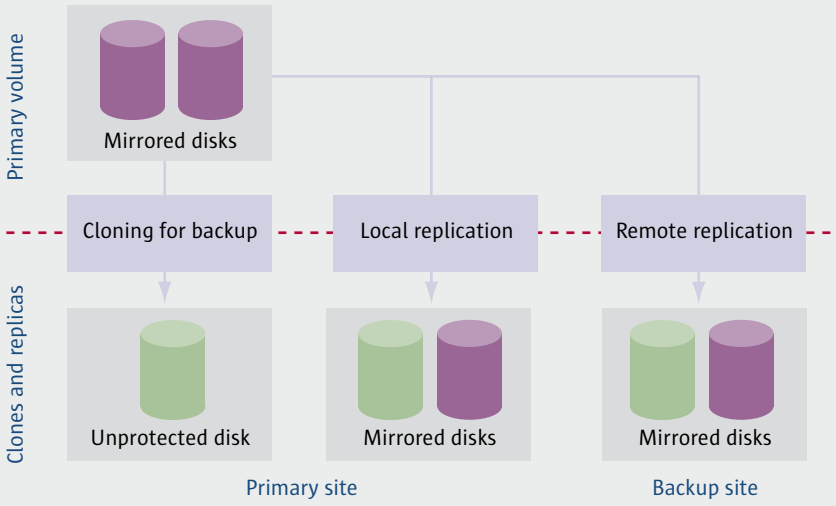
These metrics can be communicated outside the storage group and are the most appropriate to support chargeback processes, such as usage measurement and billing. Besides, the raw disk capacity metric stems from reports at the device level. It is typically more trustworthy than other metrics and definitely more reliable than reports generated from other sources, such as agents residing on servers.

3. Implement a transparent tiered chargeback for storage based on the recommended metrics and tied to an overall IT services and cost allocation model. This shapes the demand for storage technologies according to business requirements and affordability.

The lack of (appropriate) chargeback processes for storage in organizations is the source of multiple capacity issues, such as low utilization of file system capacity, configured yet unmapped capacity and inappropriate use of storage tiers. We recommend:

- Basing the model on the allocated capacity, regardless of the actual utilization. This creates an incentive for users to improve utilization of file system capacity and request only what they need and will shift the cost of unused capacity from the storage group to the user.
- Offering users options through a tiered model that considers the cost of different platforms. This provides users an incentive to request appropriate storage tiers instead of systematically demanding the most redundant configuration on high-end platforms.
- Using the raw disk capacity metric in the model to factor in the cost of disk protection (e.g., mirroring and different RAID levels) and replication. This provides users an incentive to request an appropriate degree of redundancy.

Figure 1. Configuration example (capacity efficiency ratio of 7 for critical mirrored data replicated locally and remotely and cloned for backup purposes)



Capacity efficiency ratio of this configuration: 7

Total raw capacity used = 7 times the capacity available to the file system

4. Control the use of protection and local and remote replication, or architect and fine-tune your storage environment to reduce the need for extensive protection and replication of capacity. In general, data is often overly protected and replicated without much consideration for how critical it or the business processes it supports are.

The capacity efficiency ratio, which reflects the degree of redundancy in a particular configuration or as a whole, is calculated by dividing the total raw disk capacity configured by the file system by the capacity needed to support that configuration. Figure 1 depicts a capacity efficiency ratio of 7, meaning that, by providing 1 gigabyte (GB) of file system capacity available to servers, the depicted configuration requires 7 GB of raw disk capacity.

Capacity efficiency ratios of more than 6 are not uncommon and can only be justified for critical applications with aggressive recovery objectives. A mirrored disk that is replicated locally and remotely with the same protection levels and cloned daily results in a degree of redundancy of 7 (and this excludes backup copies of the data).

To unlock wasted capacity, align protection and replication configurations with the actual recovery needs of the application. Capacity chargeback based on raw disk capacity is a recommended catalyst to help the business perform risk/cost analysis for each application.

To improve backup performance and limit its impact on production systems, multiple organizations typically configure copies of active production volumes (called business continuance volumes or clones) from which the backup is going to be performed (instead of backing up directly to primary disks). Such volumes require costly capacity on the same tier. To limit the need for such cloned capacity, backup performance should be improved by:

- Moving to a disk-based backup solution to expedite backup jobs.
- Implementing a storage area network (SAN) backup solution to avoid backup traffic clogging the network.

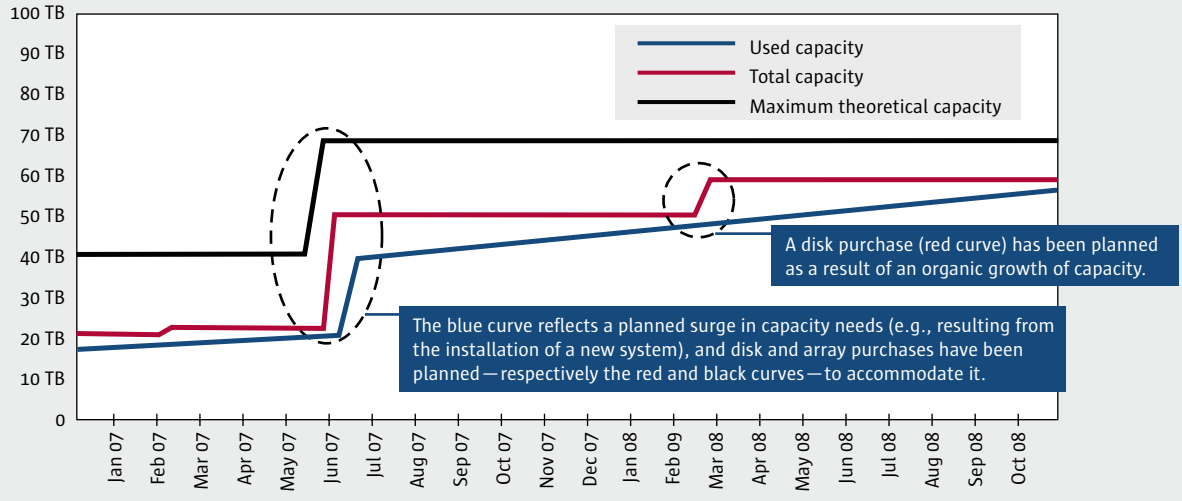
5. Consider information life cycle management (ILM) to archive information of diminishing business value automatically.

ILM is based on the paradigm that information value changes over time and this value should be used to define the processes for managing information throughout its life cycle. From a storage standpoint, this means different storage tiers should be used and automated mechanisms should be applied depending on the value of the data under consideration and its life-cycle stage.

Defining information value and ILM, in general, requires a good understanding of enterprise business processes, corporate policies, and compliance, legal and regulatory requirements. ILM techniques and discipline help in implementing a standard data classification scheme across the enterprise and, therefore, well-defined retention policies. An ILM-oriented analysis can be applied to structured or semi-structured data.

For example, e-mail is typically considered the low-hanging fruit of ILM because of the semi-structured nature of the data and the maturity of available e-mail archiving tool-

Figure 2. Snapshot of the output of BearingPoint's storage capacity planning tool



Plan for capacity: Less is more.

sets. Since the business and legal value of information is typically tied to time in the messaging world, ILM policies applying to e-mail can be fairly easily designed using that dimension.

6. Evaluate new storage virtualization technologies such as thin provisioning to help pool resources and improve overall utilization.

Storage virtualization refers to the multiple storage techniques used to create a level of abstraction that pools resources and hides the complexity of the underlying infrastructure for users and management layers. In managing capacity, thin provisioning — which consists of an automated solution to allocate capacity — can be an appealing technology. Thin provisioning essentially makes over-allocating practices irrelevant and circumvents low utilization rates by extending volumes at write-time on an as-needed basis rather than only at allocation time.

A cost-effective way to take the thin provisioning path is to leverage the capabilities of virtual disk files created by server virtualization solutions. Such virtual disks can be set up to expand dynamically to a fixed maximum size as data is saved to them. Although such a technique is not recommend for latency-sensitive data, it achieves the benefits of thin provisioning at a limited cost for organizations running virtual servers.

7. Plan for capacity: Less is more.

The most effective way to manage capacity is still to plan for it appropriately. This is the holy grail of what is commonly referred to as storage resources management (SRM), although most solutions carrying this acronym usually fail to deliver effective capacity planning capabilities. Other capacity planning solutions available are too granular or focused on the file system view (or the server view), which does not help the storage group.

Instead, storage capacity planning can be achieved simply by using simple tools (spreadsheets!) and proactive communication with the business to identify storage needs. Capturing storage requirements for each tier and location and identifying and applying the correct protection mechanisms used do not require a long and protracted process.

Overview of a planning tool

The authors have developed a storage capacity planning tool that forecasts capacity utilization and helps planning for hardware purchases across multiple storage tiers in several different locations. Capacity metrics are used at the disk level to forecast raw disk capacity usage, as well as at the array level to forecast the number of free slots in storage arrays.

This tool, an advanced Microsoft Excel® spreadsheet with macros, must be configured to the characteristics of the storage environment at hand, such as the protection and replication mechanisms used, which can be done in a matter of hours or days based on the documentation level of the current state. Once the configuration is done, the tool must be baselined with a high-level capacity snapshot of the environment. Such data can be collected straightforwardly and rapidly from existing storage reporting tools.

Once these steps have been performed, the capacity planner can schedule capacity consumption over time and plan for capacity additions (disk or frame purchases). The utilization of resources is then graphically presented for each tier of storage and each location. Maintaining the tool requires periodic re-baselining of the storage environment and updating of capacity needs resulting from the demand management process.

In a four-tier, dual-site environment of approximately 100 terabyte (TB) of managed raw storage capacity, the tool is implemented in less than three weeks, including training of the storage group. Updates, requiring only a few minutes, are scheduled biweekly and include new forecast data from the application teams and additional raw capacity purchases. Re-baselining is performed monthly in a couple of hours. The tool helps identify potential capacity shortage in specific tiers/locations and supports making informed, relevant storage purchase decisions. ■

New advances and techniques to strengthen network resiliency: A primer on new technologies

Abstract

Unpredictable issues in the global telecommunications grid can have a serious impact on business operations. Yet users take for granted that service will be always available.

Although the quality of services[1] now avail-able from telecommunications carriers is radically better than it was a mere one or two decades ago, enterprises are now completely dependant on their communication networks to function to a degree that was unimaginable in years past.

This article presents recent advances and existing technologies that have emerged as the essential building blocks for implementing added network resiliency. It also considers the budgetary considerations associated with the adoption of these techniques.

Introduction

Current events and trends underscore the unpredictability of and increasing demands on global telecommunications services. The recent failure of undersea cables near Taiwan impacted most of the international traffic in and out of Asia and seriously disrupted the flow of information that global organizations have grown to rely on to operate. In addition, managers are constantly looking for ways to improve the efficiency of their enterprise's computing infrastructure by making capacity available on demand, reducing unit cost and simplifying operations. Recent developments in utility computing[2] are helping realize some of these goals. But utility computing implementation that leverages, among other techniques, virtualization of infrastructure resources, including servers and storage, assumes the existence of a resilient data network that delivers computing utility services across the enterprise. Finally, through the convergence of voice and data on a single network, users worry about loosing the ability to communicate with coworkers at the time of a network failure.

These examples confirm how critical network resiliency has continued to be for financial services organizations. This article offers a brief primer on the solutions managers can bring to bear to implement the technology and processes that help address network resiliency challenges.

Advances in core networks

Telecommunications service providers have developed their data network services significantly, and equipment manufacturers have done so for their network equipment. Key advances the context of network resiliency and diversity include:

- Backbone networks with multiprotocol label switching (MPLS) are building on previous technologies introduced in the last decade, such as dense wavelength division multiplexing (DWDM) and synchronous optical network (SONET). These advances allow telecommunications carriers ("Carriers") to consolidate their layers 1–3[3], which operate on disparate physical networks, onto a single MPLS backbone network. Carriers can provision existing point-to-point, frame relay and ATM services, as well as new services. This consolidated backbone delivers higher reliability, differentiated level of service, ease of manageability, improved provisioning time and reduced cost of ownership. Moreover, the recently launched MPLS service (layer 3) introduced by some carriers for end users provides increased built-in resiliency and diversity compared to existing layer 1 or 2 telecommunications services.

- Routing protocols such as fast reroute (FRR), optimized edge routing (OER) and traffic engineering (TE) are helping improve network availability and performance.

Enterprises are now completely dependent on their communication networks to function to a degree that was unimaginable in years past.

[1]Expressed along any kind of dimension, including bit error rate, latency and mean time to restore.
[2]For more details, see the article, "Utility computing: Keys to successful implementation" in this issue of the BearingPoint *Financial Services Technology Journal*.
[3]Telecommunications engineers are using a conceptual view of the accumulation ("stack") of technology that must be used to deliver end-to-end services. The first three layers are physical, data and network (see http://www.ieee.org).
[4]The same decisions impact network scalability, although this discussion is out of scope of this article.

- Network equipment technology combined with configuration and software technology, such as stateful switchover (SSO) and nonstop forwarding (NSF), can significantly improve network availability during network module failure. Selecting resilient network equipment and component configurations can further reduce network downtime.

- Network equipment combined with the protocol enhancements is able to support wide area network (WAN) optimization techniques to keep bandwidth cost under control and improve application response time. Traffic optimization is implemented in WAN devices by techniques such as compression, selective ACKs, traffic reordering and limited retransmission. These optimization techniques are also able to prioritize business-critical traffic during network capacity constraints. WAN equipment with add-on modules is able to support wide area file services (WAFS) when file servers are centrally located. WAFS are helping to improve response time in remote access significantly.

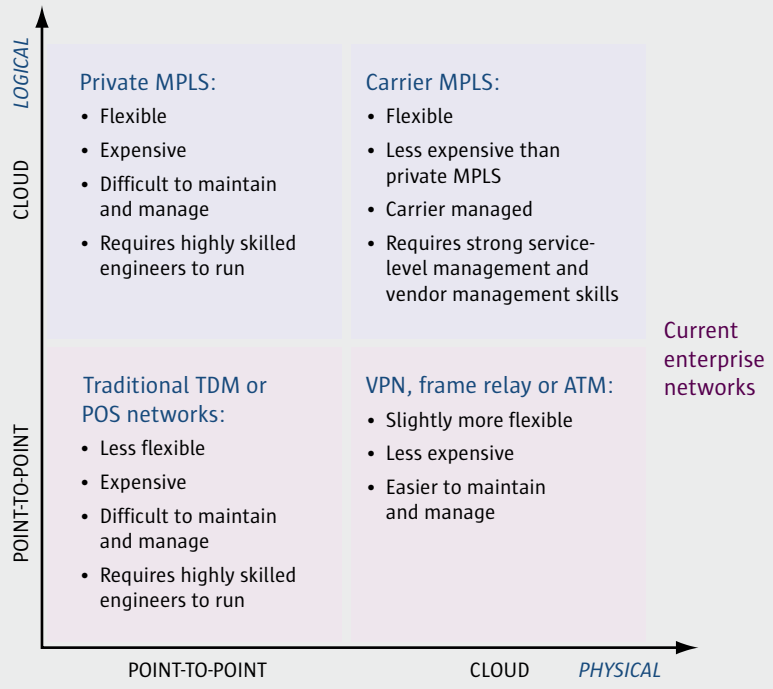
Advances in network design

A number of design decisions affect enterprise network resiliency,[4] ranging from selecting the type of telecommunications service delivered by carriers, network topology, equipment configuration, routing protocol and configuration, protocol timer tuning, telecommunications cable diversity, and point of presence (PoP) diversity.

Telecommunications services

By subscribing to layer 1 services from carriers, organizations can design, deploy and manage diversity and resiliency in their corporate network. Ownership lies with the client. On the other hand, by subscribing to a layer 3 MPLS service pro- vided by a vendor, diversity and resiliency are based on the vendor-implemented network. In certain situations, single points of failure may occur in vendor networks and network traffic may experience poor performance (increased latency). These issues may be significant based on the traffic flow and underlying connectivity of the carrier network.

Figure 1. Telecommunications services spectrum



Cost and operational differences are also quite different in various scenarios. Enterprises need to select appropriate service based on different factors, including cost, scalability, availability, manageability and in-house skills. Figure 1 provides a comparison between various telecommunications services available from carriers given how services is delivered within the enterprise ("Logical" axis) and actually supported by the carriers ("Physical" axis).

Network topology. Based on the requirements and type of telecommunication service, network backbone topology can be fully meshed or partially meshed. Depending on the level of meshing at layer 1 service, reliability improves and performance is better under failure conditions, but cost goes up with increased degree of meshing. While using a frame relay service, depending on the level of resiliency implemented, one or more permanent virtual circuits will be required. In contrast, in layer 3 MPLS service, users need not subscribe to any additional divers-ity in topology because it is inherent in the solution.

Equipment configuration. Distribution of network connectivity onto more than one device and replication of certain network modules in a device have significant impact on overall network resiliency and diversity. Replication of routing/switching, interface modules and power supply significantly improves device reliability, resulting in enhanced network availability. Distribution of network connectivity, as opposed to replication of modules, result in different recovery time and variations in latency during failure conditions based on network topology.

Routing protocol. Based on routing protocol domains overlaid onto network topology and type of routing protocol used, network convergence and recovery time can vary significantly during failure conditions. Routing protocols such as Routing Information Protocol (RIP), Enhanced Interior Gateway Routing Protocol (EIGRP), Open Shortest Path First (OSPF), Border Gateway Protocol (BGP) and FRR have different convergence times (i.e. the times it takes for the network to reconfigure after a major outbreak) when network topology changes.

Enterprises should make buy-versus-build decisions based on their networking requirements, the availability of telecommunications services, cost of services, latency and organizational maturity to manage a complex network in-house.

Telecommunication cable diversity. When designing a corporate network, cable diversity mplemented by carriers needs to be carefully studied. Depending on type of link, there are different diversity considerations, such as landing station and a long-haul international link using undersea cables. For local loop diversity, the location of the carrier's central office, cable routing and building entrances need to be analyzed.

Point-of-presence (PoP). The location of carrier PoPs are critical for supporting diversity. Even if two different carriers provide the last mile, diversity is not necessarily provided. During the design phase, PoP location needs to be carefully studied and the appropriate implementation decision made to ensure PoP diversity.

Protocol timer tuning. Each routing protocol has a number of parameters that need to be configured. The value of these parameters impact network convergence during a topology change (typically during failure conditions). The most valuable of these parameters would support faster network convergence and reduce the impact on application traffic. These timers include hold, keep alive and advertisement interval.

Implementation considerations

Once design decisions have been made, a detailed analysis is required for all of the above domains during the implementation phase. We review the most important ones below.

Telecommunications services

Enterprises should make buy-versus-build decisions based on their networking requirements, the availability of telecommunications services, cost of services, latency and organizational maturity to manage a complex network in-house, for example. The decision to buy versus build dictates whether an organization buys raw bandwidth (traditional layer 1 service) or managed bandwidth. If organizations procure raw bandwidth, they are responsible for overall manageability, diversity, resiliency and ongoing operation. Telecommunications services available are:

Layer 1:

- **T1, DS3.** Traditional point-to-point links do not offer any diversity or resiliency in the last mile from the PoP to the enterprise. Carrier backbone networks may be configured in protected or non-protected service based on the customer's subscription. To help ensure diversity in the local loop, the customer must buy at least two local loops connected to a diverse central office, routed through a diverse path from a remote site to the central office with diverse building entrances. Traffic routing on the diverse path is managed at the routing layer.
- **SONET/OC3.** This technology adds diversity in the network at the physical layer both in the local loop and the long haul and can be protected or non-protected. The idea is to use two or more transmission paths between network nodes so, if there is a break in one line, the other may still be available. For the best security against failure, when possible, different physical routes are used for the two lines. Traffic routing through diverse paths is managed at the physical layer, resulting in very fast convergence compared to network layer convergence.
- **DWDM.** DWDM is the clear winner in the backbone. It was first deployed on long-haul routes in a time of fiber scarcity. Then the equipment savings made it the solution of choice for new long-haul routes, even when ample fiber was available. While DWDM can relieve fiber exhaust in metropolitan areas, its value in this market extends beyond this single advantage. Alternatives for capacity enhancement exist, such as pulling new cable and SONET overlays, but DWDM can do more. It delivers additional value in the metropolitan market through fast and flexible provisioning of protocol- and bit rate-transparent, data-centric protected services, along with the ability to offer new and higher-speed services at a lower cost. Key requirements for DWDM systems in a metropolitan area network (MAN) include multi-protocol support, scalability, reliability and availability, and cost-effectiveness.

Layer 2:

- **Frame relay.** In last decade, frame relay has been a popular and cost-effective protocol for remote site connectivity compared to point-to-point links. Moreover, it provides the option to implement multiple permanent virtual circuits (PVCs) between two points. These PVCs offer

Figure 2. Sample template parameters

Landing station A	Landing station Z	Cable name	Actual latency	Contractual latency	Protected versus non-protected	One-time cost	Recurring cost

Global organizations have access to a range of technology and service choices to build a level of network resiliency commensurate with the strategic importance of the telecommunications infrastructure.

diversity because traffic can be routed to a second PVC if first PVC is down. However, as cost of bandwidth is decreasing and other new services such as MPLS are becoming available, organizations are phasing out use of this service. Per-unit bandwidth cost in frame relay is much more compared to new services.

- Ethernet over MPLS (EoMPLS). EoMPLS provides Ethernet service over a carrier's MPLS network. It offers quality of service features and delivered enhanced stability, scalability, bandwidth and guaranteed level of service. EoMPLS can also provide point-to-multipoint configuration. Enterprises can run their own routing protocol based on their requirements on top of this service.

Layer 3:

- MPLS. The MPLS layer adds diversity in the network by providing dynamic rerouting in case of network element failure. This mechanism, called MPLS FRR or MPLS local protection, uses backup paths in case of link failure and provides fast recovery compared to the recovery mechanisms employed at the Internet Protocol (IP) layer. Any-to-any connectivity, inherently supported by MPLS, supports diverse access to computing and resources on the network.

  Once an organization decides on telecommunications services, it then must select appropriate telecommunications cables and PoPs.

Telecommunications cables

When selecting carriers during the implementation phase, for each international link, organizations should identify the locations of landing stations at both ends and collect routing information. When two different links are engineered for resiliency purposes, their diversity is confirmed by engineering a diverse end-to-end path. A diverse path may impact link latency and cost. Therefore, when requesting a carrier's proposal, an organization should develop detailed templates and request such infor-mation from the carrier. Sample template parameters are given in Figure 2.

The template parameters are further enhanced by including details of the terrestrial portion of end-to-end circuits.

Cable routing from the local PoP or central office to the customer building should be fully analyzed. Buildings should have diverse entrances, cable risers and communication rooms. Carriers should be asked to provide the PoP address and cable routing map to complete analysis of diversity in the cable engineering.

Points of presence

PoPs are critical to support diversity in the network. Carrier networks may have diverse cabling and equipment infrastructure but may be passing through the same PoP, resulting in high risk.

For example, in a recent analysis of PoP diversity in a carrier MPLS network, a customer site was connected to two separate MPLS nodes in two different countries in the carrier network. However, during detailed analysis, it was found that the physical circuit to the second MPLS node was going through a PoP with the first MPLS node.

Routing protocols

Fast reroute (FRR). During link failure, traditional routing protocol may take less than 10 seconds to recover the network path through an alternate link in the backbone. However, if SONET links are used, FRR can establish an alternate path in less than 50 micro-seconds. As a result, real-time traffic flows are not impacted.

Optimized edge routing (OER). This protocol is used at the remote site to route traffic in load sharing mode on two different links or between primary and secondary links during link failure. OER protocol parameters should be balanced, reducing both network convergence time and protocol processing overhead.

Equipment configuration

Network software resiliency. Backbone routers should take advantage of protocol redundancy features such as SSO/NSF when outfitted with redundant route/switch processors. SSO/NSF supports continuous routing of traffic during certain hardware failures.

Conclusion

Enormous pressure has always been put on managers of the telecommunications infrastructure to deliver network connectivity with a high level of predicable quality. Fortunately, global organizations have access to a range of technology and service choices to build a level of network resiliency commensurate with the strategic importance of the telecommunications infrastructure.

Added network resiliency comes at the cost of building redundancy at various levels (for example, at the equipment, circuit, central office, backbone/PoP, site, regional and carrier levels), as well as other mitigating technologies and design patterns presented in this article. Accordingly, the need to develop a business case for implementation is just as critical as the need to select and implement any one of the available solutions. This discussion is out of the scope of this article. But, in our experience, no organization can ignore the need to design and maintain a strong network infrastructure to support a credible enterprise survival plan. The question becomes: When global operations are affected by network failure, how much would you give to makes service resume then and there? ■

About the authors

Manuel Barbero is the practice leader for the Financial Services Technology practice in both the Financial Services and Cross-Industry solutions business units of BearingPoint and serves as advisor to BearingPoint's executive committee as Chief Technology Officer for client service. He is the executive editor of the *Financial Services Technology Journal* and the panel chair for BearingPoint's Innovation Program.

Daryouche Behboudi is a managing director with BearingPoint's Financial Services Technology practice. He has managed consulting engagements with top-tier financial services firms, focusing on enterprise network architecture, IT cost takeout, IT strategy articulation, IT operations and organizational design, IT procurement and outsourcing, and project management office and implementation.

Julien Furioli is a manager in the Financial Services Technology practice of BearingPoint. Julien is responsible for a wide range of solutions in areas including IT infrastructure sourcing, market data infrastructure, and trading applications support infrastructure and instrumentation.

Rajiv Jain is a senior manager in BearingPoint's Financial Services Technology Solutions practice. He has more than 20 years' IT experience in a wide range of technologies spanning global communication networks to core computing systems and architectures. He assists financial institutions in deploying state of the art infrastructures globally to support the launch of new services and products, reduce costs, and manage existing environments.

Gabriel Leung is a senior manager in BearingPoint's Infrastructure Solutions practice. He has more than 15 years' experience serving in senior management capacities including chief information officer and chief technology officer. His focus areas include strategy development, enterprise architecture design, high-availability data center design, utility computing, communication technologies and IT security. He works with clients in industries including banking and financial services, manufacturing, energy, software product, and consulting.

Arnaud Merlet is a senior consultant in the Financial Services—Technology Infrastructure and Integration practice of BearingPoint. He is involved in several utility computing, data center, and storage and data life cycle initiatives and has an experience spanning the functional areas of service catalog and chargeback, infrastructure assessment, and policies and procedures development.

Anne Mondesert is a management analyst in the Financial Services Technology practice of BearingPoint. Anne has played an active role in IT sourcing strategy and cost optimization engagements.

Zahra Rahemtulla is a manager in the Financial Services—Technology Infrastructure and Integration practice of BearingPoint. She has extensive experience in managing, operating and delivering infrastructure solutions across many industry sectors. Zahra holds the ITIL Service Manager certification and has assisted several global financial services organizations in the design and implementation of service management solutions.

Pinaki Roy is a senior manager in BearingPoint's Financial Services Infrastructure Solutions practice. He has more than 10 years' experience in a wide range of technologies including Web-based application development, infrastructure design, implementation and support. He assists financial institutions in designing and deploying solutions to facilitate new business opportunities, control costs and better manage existing environments. He serves as the technical lead for BearingPoint's utility computing solution.

Laurence Thiery is a senior manager in the Financial Services Technology practice of BearingPoint. She is the deputy editor of the Financial Services Technology Journal and a member of the BearingPoint's Innovation Program selection committee. She has over 10 years of experience in system integration and infrastructure projects.

Frederic Véron leads BearingPoint's Financial Services Infrastructure Solutions practice. He has more than 15 years' experience in a wide range of technologies spanning global communication networks and core computing systems and architectures. He assists financial institutions in deploying modern infrastructures on a worldwide basis to support new business models, better control costs and manage their existing environment. He also leads BearingPoint's IT Strategy & Transformation Solution in the Financial Services group.

Takashi Yoshizaki is a manager in BearingPoint's Financial Services practice. He is involved in projects covering a wide range of technologies and their application to business processes including distributed computing systems, data/voice/voice over Internet protocol (VOIP) networks, data centers, and asset management and security systems.

Previous issues of BearingPoint's *Financial Services Technology Journal* are available at www.bearingpoint.com/fstechjournal. To contact the editor, send an e-mail to manuel.barbero@bearingpoint.com.

MAY 15, 2004 | Volume 1 | Issue 1

Letter From the Executive Editor

The Business of Technology	Enabling the Holistic Enterprise With Web Services and Service-Oriented Architecture
How It (Should) All Start	Fighting Babel With an Architecture Notation Guide
Data For Decision	IP Telephony: The Enterprise Business Case
Below The Code	Data Center Rationalization: Why and How

DECEMBER 6, 2004 | Volume 1 | Issue 2 DIGITIZATION IN FINANCIAL SERVICES

Letter From the Executive Editor

The Business of Technology	Digitization: The Time is Here Enterprise Credit Management: Digitization Comes of Age
How It (Should) All Start	Business Process Management: Taking All the Right Steps Document Generation: Digitization in Action Enterprise Information Integration: Bridging the Digital Divide
Data For Decision	Digitization and Security in Financial Services

SPRING 2005 | Volume 2 | Issue 1 THE RISK, COMPLIANCE AND SECURITY ISSUE

Letter From the Executive Editor

This Issue at a Glance

The Business of Technology	How to Create a Holistic Risk and Compliance Framework Part I: Building the Data and Workflow Infrastructure Part II: Integrating Systems, Processes and Tools
How It (Should) All Start	Risk and Identity Management for Financial Services Designing and Implementing a Continuity of Operations Solution for Financial Services Firms
Data for Decision	Using Real Options to Mitigate Risks in Large IT Projects Monte Carlo Analysis for Technology Project Risk Modeling
Below the Code	Data Security, Privacy, Integrity and Compliance Infrastructure Implications of the Convergence of Risk and Compliance

SPRING 2006 | Volume 3 | Issue 1 THE STRATEGY AND ARCHITECTURE ISSUE Part One: Concepts and Techniques

Letter From the Executive Editor

This Issue at a Glance

The Business of Technology	The Renaissance of Enterprise IT Architecture Connecting Business and Technology Strategies Through Enterprise IT Architecture
How It (Should) All Start	Architecting the IT Organization
Data for Decision	The Role of Patterns in Enterprise Architecture Development
Below the Code	Enterprise Architecture Notation—Let's Get Physical Myths and Realities of Open Source

FALL 2006 | Volume 3 | Issue 2 THE STRATEGY AND ARCHITECTURE ISSUE Part Two: Architecture in Motion

Letter From the Executive Editor

This Issue at a Glance

The Business of Technology	Telecom Expense Management Solution Architecture
How It (Should) All Start	21st Century Messaging Compliance Architecture: E-mail Archival for Large Regulated Enterprises Call Center Virtualization Solution Architecture
Data for Decision	Rule Service Architecture
Below the Code	Implementing an Application Architecture With Open Source Implementing a Network Architecture

BearingPoint. Management & Technology Consultants

Helping our clients get sustainable, measurable results

BearingPoint is a leading management and technology consulting company serving the *Forbes* Global 2000 and many of the world's largest public services organizations. Our more than 17,000 passionate, experienced consultants help organizations around the world solve their most pressing challenges, day in and day out. Through our collaborative and flexible approach, we help our clients get practical, sustainable, measurable results, make the right strategic decisions and implement the right solutions. We are BearingPoint, management and technology consultants.

To learn more, contact us at 1 866 BRNGPNT (+1 508 216 2523 from outside the United States and Canada), or visit our Web site at www.bearingpoint.com.

BearingPoint, Inc.
1676 International Drive
McLean, VA 22102

www.bearingpoint.com

© 2007 BearingPoint, Inc. All rights reserved. Printed in the United States. BearingPoint® is a registered trademark of BearingPoint, Inc. or its affiliates in the United States and other countries. Any other marks are the property of their respective owners. C3937-0607-01-USRD???